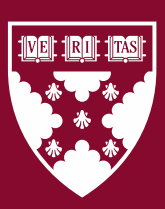
Working Paper 26-054

The Equity Market Implications of the Retail Investment Boom

Philippe van der Beck
Cameron Cohen
Coralie Jaunin



Harvard
Business
School

The Equity Market Implications of the Retail Investment Boom

Philippe van der Beck
Harvard Business School

Cameron Cohen
Harvard Business School

Coralie Jaunin
Université de Lausanne & Swiss Finance Institute

Working Paper 26-054

Copyright © 2026 by Philippe van der Beck, Cameron Cohen, and Coralie Jaunin.

Working papers are in draft form. This working paper is distributed for purposes of comment and discussion only. It may not be reproduced without permission of the copyright holder. Copies of working papers are available from the author.

Funding for this research was provided in part by Harvard Business School.

The Equity Market Implications of the Retail Investment Boom*

Philippe van der Beck† Cameron Cohen‡ Coralie Jaunin§

Abstract

We quantify the impact of Robinhood traders on the US equity market. Within a structural model of asset demand, we estimate investor-specific demand curves for retail and institutional investors and derive aggregate pricing implications from market clearing. Robinhood traders exhibit substantially different demand curves from all other investors, including households. Their demand is the most elastic, their portfolios tilt most strongly toward volatile and micro-cap stocks, and their latent demand responds sharply to large absolute recent returns, reflecting attention-driven trading. While their aggregate impact is small on a value-weighted basis, per dollar of invested wealth Robinhood traders exert the largest price impact of any investor group, including hedge funds. Their high marginal impact raises concerns about the surge in Robinhood assets and mobile-first retail brokerages more broadly: during the 2020 lockdown, Robinhood trades explained over 13% of the variation in small-cap returns.

*We thank Pierre Collin-Dufresne, Yufeng Wu, Semyon Malamud, Donald Keim, Malcolm Baker, Eric Jondeau, Michael Rockinger, and the seminar participants at the University of Lausanne, École Polytechnique Fédérale de Lausanne, the Wharton School of the University of Pennsylvania, Northern Finance Association Meeting 2022, CFM Research Seminar, SFI Research days, 14th International Risk Forum, and the Society of Quantitative Analysis for helpful comments. Philippe van der Beck acknowledges financial support from the Swiss National Science Foundation (SNF).

†pvanderbeck@hbs.edu, Harvard Business School

‡ccohen@hbs.edu, Harvard Business School

§coralie.jaunin@unil.ch, Université de Lausanne & Swiss Finance Institute

1 Introduction

The rapid growth of commission-free brokerage platforms has substantially altered the investment landscape for retail investors. Robinhood, which pioneered zero-commission trading, has expanded its assets under management from $14B to over $300B in the past five years, a period in which retail trading doubled to nearly 20% of total equity volume in the US.[1] Periods of elevated Robinhood trading activity have often coincided with sharp increases in equity valuations, raising questions about the broader market impact of retail investors. High-profile events, such as with Hertz in May 2020,[2] GameStop in January 2021 and May 2024,[3] and Beyond Meat in October 2025[4] have fueled speculation that coordinated or concentrated retail demand can substantially impact market efficiency and volatility. While these events attracted significant media attention, the broader role of retail investors in shaping equity prices remains less well understood.

Motivated by the surge in retail trading activity, we seek to quantify the impact of retail traders' demand on the US equity market within a structural model of asset demand. While individual elements of retail traders' demand curves have been studied extensively, including demand for hard-to-value stocks (Laarits and Sammon, 2024), attention-driven trading (Barber et al., 2022b), or liquidity provision (Kaniel et al., 2008), assessing the equilibrium impact of these preferences on asset markets requires a structural model that

[1] See `https://www.barrons.com/articles/robinhood-stock-retail-trading-volume-91334d59` and Robinhood's "Monthly Metrics" (`https://investors.robinhood.com/financials/monthly-metrics`)

[2] Shortly after it filed for bankruptcy on May 22nd, 140,000 Robinhood traders added Hertz to their portfolios. Hertz's share price simultaneously jumped by over 1,000%.

[3] In early 2021, retail traders coordinated on the subreddit "r/wallstreetbets" to purchase shares and call options of stocks they deemed undervalued. The share price of Gamestop rose from $17.25 per share on January 4th to over $500 per share on January 28th. In May 2024, the stock surged again after the social media finance influencer "Roaring Kitty" re-appeared on X.com.

[4] In October 2025, retail traders hit record levels of single-stock trading volume, elevating the price of Beyond Meat more than 1,300% in four days.

jointly matches retail *and* institutional demand. We estimate a demand system along the lines of Koijen et al. (2024) to quantify the equilibrium pricing implications of the retail investment boom. The demand system accounts for the endogeneity of demand and prices, and is therefore able to elicit the *equilibrium* effect of the surging retail demand on asset prices.

We approximate the Robinhood portfolio using a dataset of Robinhood account holdings collected by Robintrack from 2018 through 2020. Institutional portfolios are obtained from 13F filings. The residual shares not held by 13F institutions or Robinhood traders make up the household portfolio as in Koijen and Yogo (2019) (hereafter KY). While we do not observe the exact dollar holdings of Robinhood investors, we approximate their portfolio based on the number of accounts invested in each stock, a statistic validated in Welch (2022). We further validate this approximation in a series of robustness tests. We estimate characteristics-based demand curves for institutional investors, households, and Robinhood traders over the sample from 2018 through 2020, accounting for the endogeneity of prices and latent demand. Importantly, while portfolio weights are approximated, we obtain exact figures of Robinhood traders' total asset holdings from Robinhood's SEC filings.

Robinhood traders' demand curve differs sharply from that of all other investors. While the average investor demand for systematic risk is negative, Robinhood traders exhibit extremely high and positive demand for risky stocks—more than four times that of hedge funds. Similarly, whereas the average investor has a positive demand for book equity, Robinhood traders' demand for book equity is strongly negative, consistent with their preference for micro-cap stocks. They are also the most elastic investor group, being eight times more price elastic than institutions and four times more elastic than other households. Perhaps most strikingly, their preferences for stock-specific characteristics diverge from those of other

households, with coefficients consistently carrying the opposite sign. This highlights that this novel class of retail trading behavior is fundamentally distinct from conventional household portfolio choice (Balasubramaniam et al., 2023; Gabaix et al., 2025). We furthermore show that the estimated demand coefficients of Robinhood traders are remarkably stable over time, in line with the persistent habitats of retail traders documented in Laarits and Sammon (2024).

Based on the estimated demand curves and Robinhood traders' assets under management (AUM) disclosed in Robinhood's SEC filings, we impose market clearing and derive equilibrium equity prices as a function of the estimated demand coefficients, AUM of institutions, households, and Robinhood traders, and stock-specific fundamentals. We use the estimated demand system to quantify the marginal impact of each investor type on the cross-section of stock prices. Following Koijen et al. (2024), we sequentially redistribute the assets of Robinhood traders, households, and institutional investors to the market portfolio and compute the total repricing of equities relative to each group's market share. This exercise yields the marginal price impact of an invested dollar for each investor type. In equilibrium, this marginal impact depends on how much a group's demand curve deviates from the asset-weighted average demand curve of other investors, as well as on its own and others' price elasticities of demand. Strikingly, Robinhood traders exhibit the highest impact per dollar of invested wealth among *all* investor groups. Their repricing per invested dollar is \$6.7, compared to \$4.5 for hedge funds and \$2.7 for other households. The high marginal impact reflects their strong preference for micro-cap and high-risk stocks, as well as their highly elastic and attention-driven demand, all of which deviate sharply from the rest of the market. We also consider alternative counterfactuals by redistributing Robinhood traders' assets across other investor groups such as households or investment advisors and find similar

magnitudes. While Greenwood et al. (2023) and Li (2025) demonstrate that retail traders can have a sizable effect on stock markets, we show that their marginal impact exceeds that of all other investor groups. This finding indicates that the rise of mobile trading has distinct implications for the cross-section of equity prices, fundamentally different from traditional household demand. Given their high marginal impact, the continued expansion of retail trading is likely to play an increasingly important role in shaping asset prices (Asness, 2024).

Next, we turn to the impact of Robinhood traders on the volatility of stock returns. While Robinhood traders' *marginal* impact per invested dollar is high, their total impact on volatility depends on the scale of their invested capital, as well as the magnitude of *changes* in their demand curve. To quantify their aggregate impact on volatility, we turn to 2020, when retail trading activity—presumably fueled by commission-free brokerage, stimulus checks, and limited consumption opportunities—increased sharply. During 2020, Robinhood traders' total equities held quadrupled from \$13.5 to \$53 billion. We decompose the cross-sectional variation in stock returns into supply- and demand-driven components and find that the latent demand of Robinhood traders accounts for roughly 13 percent of the return variation among small-cap stocks (bottom quintile of market equity). In contrast, their impact on large-caps (top quintile) is effectively zero. This finding underscores that retail trading activity is primarily relevant for understanding the volatility of small stocks. However, as retail participation continues expanding, its influence could increasingly extend to larger stocks as well.

Motivated by the role of Robinhood traders' latent demand in explaining the volatility of small stocks, we examine the determinants of this latent demand. Consistent with Welch (2022), we find that Robinhood traders' latent demand responds strongly to both recent trading volume and large absolute past returns. Notably, their latent demand with respect

to returns exhibits a distinct V-shaped pattern. This tendency to chase both positive and negative returns aligns with attention-induced herding behavior (Barber and Odean, 2008; Pedersen, 2022). These two components, recent trading volume and absolute returns, together explain nearly 25% of Robinhood traders' latent demand. In contrast, neither institutional investors nor other households display comparable attention-driven trading behavior.

Their responsiveness to large *absolute* returns implies contrarian behavior toward stocks with recent negative return shocks and momentum trading in stocks with recent positive return shocks. To test this hypothesis, in the final part of the paper we assess the return implications of Robinhood's participation on individual stocks during the market crash and recovery in the first half of 2020. We show that for the stocks with the largest falls in value during the first quarter, Robinhood traders provided liquidity, whereas for the stocks with the largest rallies in the second quarter, Robinhood traders magnified these effects substantially. To further unpack the price implications at the individual stock level, we show that Robinhood traders' price impact can be approximated as a product of their ownership share and an inelasticity multiplier, determined for each stock by the ownership-weighted demand elasticity of its investor base. Despite holding a relatively small fraction of total shares outstanding, Robinhood traders were most influential on the valuations of these stocks with high multipliers, as they were primarily held by large passive investors with strongly inelastic demand curves.

Our paper contributes to the growing literature on demand-based asset pricing that uses investor holdings to estimate investor-specific demand elasticities and identify their impact on asset prices (Koijen and Yogo, 2019; Gabaix and Koijen, 2021; Chaudhry and Li, 2025; Huebner, 2023; Li et al., 2024; Jiang et al., 2024; Bretscher et al., 2025; Fang et al., 2025; Jansen, 2025). We follow the methodology of Koijen and Yogo (2019) and Koijen et al. (2024),

but split the household sector into Robinhood traders and ordinary households. Robinhood traders have fundamentally different demand curves from other investors and their latent demand is – unlike any other investor group – well-explained by demand for past positive and negative returns. This suggests that these retail traders' demand shocks are to a large extent driven by attention-induced trading (Barber et al. (2022a)). In line with the recent demand-system literature and the reduced-form evidence on index reconstitutions (Shleifer, 1986; Chang et al., 2015; Pavlova and Sikorskaya, 2023), we find that institutional investors – who hold roughly 65% of the US equity market – exhibit very low price elasticity. In fact, 40% of all invested wealth is held by passive institutional investors that are fully inelastic. This inelasticity of equity markets may reflect strict investment mandates (Gabaix and Koijen, 2021), a limited pass-through from prices to expected returns (Davis et al., 2023), or the absence of close substitutes for individual stocks (Wurgler and Zhuravskaya, 2002). Because institutional investors are inelastic and Robinhood traders' preferences deviate sharply from the market average, their demand generate disproportionately large repricing effects.

Second, our paper contributes to the analysis of retail investors' demand and their behavioral biases. Recent work by Laarits and Sammon (2024) shows that behavioral biases such as extrapolation (Benartzi and Thaler, 2001), attention (Barber et al., 2022b), familiarity (Huberman, 2001; Seasholes and Zhu, 2010), and the disposition effect (Odean, 1998; Dhar and Zhu, 2006; Vaarmets et al., 2019) lead to persistent "retail habitats." We build on this literature by moving from descriptive evidence of retail trading behavior to a *structural estimation of investor demand curves*. This shift offers three advantages.

First, it allows us to separate retail investors' demand *shifts* from movements *along* their demand curve, i.e., their reactions to prices. Retail traders emerge as the most elastic investor group, typically providing liquidity through price-sensitive trading, yet simultaneously

exhibiting large, coordinated shifts toward small, risky, and attention-grabbing stocks.

Second, it enables direct comparison of retail preferences with those of other investor groups, providing a benchmark for how *excessive* certain behavioral patterns are. Robinhood traders, in particular, display fundamentally different demand from the aggregate household sector (Koijen and Yogo, 2019), with extreme preferences for volatile and micro-cap stocks and unusually high trading activity in individual names. While Gabaix et al. (2025) document low trading activity by households in the aggregate stock market, our findings concern intensive trading in individual stocks, reflecting the behavior of a small subset of active retail investors. This pattern is consistent with the excess turnover reported in Barber and Odean (2000a, 2002).

Third, the structural perspective allows us to quantify both the marginal and aggregate price impact of the retail investment boom, extending the behavioral evidence in prior work to equilibrium asset-pricing implications.

Our findings help clarify two competing narratives about retail investors. One perspective emphasizes retail traders as liquidity takers, concentrating in popular stocks and moving prices through correlated shifts in demand (Barber and Odean, 2008; Barber et al., 2008; Greenwood and Nagel, 2009). Barber et al. (2022a) and Eaton et al. (2022) document that this "herding" behavior is particularly prominent for Robinhood traders and so they may increase market makers' inventory risk. Meanwhile, a long literature has shown that retail traders act as contrarians and liquidity providers (Kaniel et al., 2008, 2012; Kelley and Tetlock, 2013; Barrot et al., 2016). After institutions sold off heavily in early 2020, several studies find that Robinhood traders stepped in as liquidity providers (Pagano et al., 2021; Welch, 2022; Glossner et al., 2025).[5] By disentangling the slope of retail demand (liquidity

[5]In a complementary paper, Ozik et al. (2021) find that illiquidity attenuated during lockdowns. However, they find that the increased liquidity is smaller for "attention-grabbing" stocks, which is also consistent with

provision) from the volatility and direction of retail demand shifts (liquidity taking), our estimates help reconcile both accounts: retail traders are highly elastic, thereby contributing to liquidity provision, yet their concentrated demand shocks can have large price effects in markets dominated by inelastic institutional investors.

The remainder of this paper is structured as follows. Section 2 describes the stock market and holdings data. Section 3 describes the demand system and the estimation procedure. Section 4 presents our empirical results on the impact of Robinhood traders on the equity market. Section 5 investigates the determinants of Robinhood traders' latent demand. Section 6 concludes.

2 Data

2.1 Stock Characteristics

Our initial sample includes all common shares traded on NYSE, AMEX, and NASDAQ between January 2000 and June 2020. Accounting and financial data are obtained from Refinitiv. Following KY (2019), we use log market equity, log book equity, profitability, investment, dividends to book equity, and market beta as stock-specific characteristics in the demand estimation.[6] We construct an outside asset from the set of firms either not headquartered in the US, real estate investment trusts (GICS classification 601010), or firms with missing characteristics. Our final investment universe contains 2,539 stocks, of which 2,376 are held during the primary sample in 2018-2020.

the herding view.

[6]See KY (2019) for further details on data construction.

2.2 Institutional Holdings

Holdings data from Securities and Exchange Commission (SEC) 13F filings are obtained from Refinitiv Ownership. Based on Refinitiv's investor labels, we group institutions into six types: banks, pension funds, insurance companies, investment advisors, hedge funds, and others. In the construction of institutional portfolios, we follow KY (2019). Further description of the ownership construction is available in Appendix B. We compute the percentage of ownership as the number of shares held divided by shares outstanding. Finally, we construct each institution's equity portfolio weights as the dollar holdings in each stock (price times shares held) divided by their AUM. We compute assets under management as the sum of an institution's dollar holdings.

2.3 Robinhood Holdings and Other Household Holdings

While Robinhood does not disclose the actual number of shares held by its user base, the number of registered accounts invested in a particular stock over time is publicly available and has been compiled on Robintrack from March 2018 to August 2020.[7] We start from the Robintrack dataset and exclude holdings outside our investment universe. Let $\text{Acc}_t(n)$ denote the number of Robinhood accounts holding stock n at time t. Figure 1 plots the average number of Robinhood accounts holding a stock, i.e., $\frac{1}{N}\sum_{n=1}^{N}\text{Acc}_t(n)$, from 2018 through 2020. The figure visualizes the surge in retail trading activity during the onset of COVID, when the S&P 500 plunged.

[7]The website `https://robintrack.net/` relied on Robinhood's own API to obtain the counts of open stock-account positions on the platform. Since Robinhood does not allow its users to short sell, this number solely reflects long positions. Options are also excluded from the count. The API was decommissioned in August 2020.

Figure 1: **Evolution of Robinhood Account Holdings**
The left axis of the figure plots the daily average number of Robinhood accounts holding a stock within our investment universe. The number of accounts holding a stock $\text{Acc}_t(n)$ was directly reported on Robinhood and compiled through August 2020 by Robintrack. The right axis plots the level of the S&P500 over time.



Table A.1 reports further summary statistics on $\text{Acc}_t(n)$ during our sample. Between June 2018 and June 2020 the number of open stock-account positions on Robinhood, i.e., $\sum_{n=1}^{N} \text{Acc}_t(n)$, grew from 4 million to 23 million. On May 4th, 2020, Robinhood announced that it currently had 13 million users and had signed 3 million new users in the first quarter alone. These numbers suggest that the boom in retail trading manifested itself in both the entry of new retail traders and an increase in their average portfolio size. The cross-sectional distribution of $\text{Acc}_t(n)$ is strongly (positively) skewed. In June 2020, the majority of stocks were held by less than 1,000 Robinhood accounts, while some exceptionally popular stocks such as Ford and General Electric were held by more than 500,000 users.

While we do not observe the exact shares held by each Robinhood user, we follow Welch (2022) and approximate for their portfolio weights using $\text{Acc}_t(n)$. Let $\mathbb{1}_{a,t}(n)$ denote an indicator function equal to 1 if Robinhood account a holds stock n at time t. The number of accounts holding n at time t is then given by $\text{Acc}_t(n) = \sum_{a=1}^{\text{Acc}_t} \mathbb{1}_{a,t}(n)$ where Acc_t is the total number of registered Robinhood users. Cross-sectional differences in $\text{Acc}_t(n)$ indicate which stocks are relatively more preferred by Robinhood traders. We then define the representative Robinhood portfolio weight in stock n at time t as

$$w_t^{\text{RH}}(n) = \frac{\text{Acc}_t(n)}{\sum_{n=1}^{N} \text{Acc}_t(n)}. \tag{1}$$

This representative portfolio is also used in Welch (2022), who validates it as an empirically sound proxy for retail holdings in prior data. For example, Welch finds that this "crowd wisdom" portfolio, when constructed on portfolio data from Barber and Odean (2000b), was 97.1% correlated with the true equal-weighted household portfolio. Welch (2022) also interprets this representative portfolio as a monotonic transformation of the "breadth of ownership" measure from Chen et al. (2001).[8] We follow the literature in using this measure as a representative, though possibly noisy, proxy for retail portfolio weights. In the next section, we explore alternative definitions of Robinhood portfolios and verify that our main findings are robust to these variations.

Beginning with Robinhood's SEC S-1 filing on July 1, 2021, in anticipation of its IPO, and in every subsequent quarterly 10-Q, the firm has disclosed the exact number of users and total platform assets at the end of each quarter. Over the reporting period from December 2019 to July 2025, Robinhood's user base expanded from 7.2 million to 26.5 million, while

[8]To see this, define breadth as $b_{n,t} = \text{Acc}_t(n)/\text{Acc}_t$. Since $\sum_{n=1}^{N} \text{Acc}_t(n) = \text{Acc}_t \bar{H}_t$, with $\bar{H}_t$ the average number of holdings per investor, it follows that $w_t^{RH}(n) = \frac{\text{Acc}_t(n)}{\text{Acc}_t \bar{H}_t} = \frac{1}{\bar{H}_t} b_{n,t}$.

users' total equity holdings increased more than tenfold from $13.5 billion to $160 billion.[9] Using these AUM figures in conjunction with our portfolio approximation (1), we infer the shareholdings of Robinhood traders (see Appendix A for details).

We follow KY (2019) and construct a household sector as the residual shares outstanding not held by 13F institutions or Robinhood traders. This sector also absorbs institutions with less than $10 million under management or without positions in the inside and/or outside assets, and therefore includes households, small asset managers, and other non-13F institutions. The total assets of the household sector are constructed as the sum of their implied dollar holdings.

2.4 Alternative Robinhood Portfolios

We acknowledge that the approximation of Robinhood traders' portfolio weights may affect (i) their estimated demand coefficients, (ii) the inferred household portfolio, and (iii) our equilibrium price-impact estimates. For example, equation (1) may understate weights in large stocks and overstate them in small stocks if wealthier accounts tilt towards market weights. To address this concern, we estimate alternative demand curves under different portfolio approximations. We first compute a price-weighted Robinhood portfolio

$$w_t^{\text{RH,prc}}(n) = \frac{\text{Acc}_t(n)P_t(n)}{\sum_{n=1}^{N} \text{Acc}_t(n)P_t(n)}. \tag{2}$$

The price-weighted Robinhood portfolio correctly captures their aggregate holdings if every account holding $\text{Acc}_t(n)$ on Robinhood is proportional to the number of shares purchased (as

[9]Starting in Q1 2025, Robinhood only reports Total Platform Assets instead of Assets Under Custody (AUC), which includes assets not custodied by Robinhood. The $160 billion figure may therefore include some assets held by other institutions, although Robinhood's last reported AUC from Q4 2024 was $131 billion. All reported values used to infer Robinhood's assets during our sample include only assets for which Robinhood was the custodian.

opposed to the number of invested dollars). Next, we consider two alternative transformations of account holdings into weights based on institutional breadth of ownership. To this end, we estimate the relationship between the number of institutions holding a stock $\text{Inst}_t(n)$ and their aggregate (value-weighted) portfolio weight $w_t^{\text{Inst}}(n)$. The raw correlation between $\text{Inst}_t(n)$ and $w_t^{\text{Inst}}(n)$ is already 76% (Spearman's rank correlation $\rho = 92\%$). This implies that the number of holders can be used as a reasonable proxy for the aggregate portfolio of an investor group. Fitting a third-order polynomial to the relationship increases the correlation to 92%. Figure A.1 in the Appendix graphically illustrates the strength of the fit. Similarly, fitting an exponential function to the institutional portfolio weights increases the correlation to 88%.[10] We use each of these estimated mapping functions $f_{\text{poly}}(\cdot)$ and $f_{\exp}(\cdot)$ to transform $\text{Acc}_t(n)$ into Robinhood portfolio weights.

$$w_t^{\text{RH,poly}}(n) = \frac{f_{\text{poly}}(\text{Acc}_t(n))}{\sum_{m=1}^{N} f_{\text{poly}}(\text{Acc}_t(m))} \qquad w_t^{\text{RH,exp}}(n) = \frac{f_{\exp}(\text{Acc}_t(n))}{\sum_{m=1}^{N} f_{\exp}(\text{Acc}_t(m))} \tag{3}$$

Appendix Table A.2 reports the correlations across the alternative Robinhood portfolios. The price-weighted, polynomial-weighted, and exponentially-weighted portfolios are 32%, 76%, and 75% correlated with the baseline portfolio, respectively. Although these portfolios differ in construction and are therefore not perfectly aligned, their average tilts, and the consequences of these tilts, are highly aligned and not dependent on any of the particular specifications. Section 3.2.D shows that while the precise coefficients in the estimated demand curve vary across specifications, the main qualitative results remain unchanged. Across all four constructions of the Robinhood portfolio, Robinhood traders have the highest marginal impact per dollar wealth amongst all institution types, had a substantial effect on the volatility

[10] Specifically, we fit $w_t^{\text{Inst}}(n) = \alpha + \sum_{k=1}^{3} \beta_k (\text{Inst}_t(n))^k + \epsilon_t(n)$ as well as $\log w_t^{\text{Inst}}(n) = \alpha + \beta \log \text{Inst}_t(n) + \epsilon_t(n)$, and denote the mapping functions as $f_{\text{poly}}(\cdot)$ and $f_{\exp}(\cdot)$ respectively.

of small cap stocks, and are the only investor group with significant attention-induced demand (i.e., a strong preference for high past absolute returns and high recent trading volume).

3 A Demand System Approach to the Impact of Retail Traders

We adapt the demand system framework from Koijen and Yogo (2019) and Koijen et al. (2024) to quantify the impact of the growth of Robinhood investors on the US equity market. To this end, we decompose the household sector into Robinhood traders and the residual, which we refer to as other households, as described above.

3.1 The Model

Adopting notation from KY (2019), consider N assets over T periods, indexed by $n = 1, ..., N$ and $t = 1, ..., T$. Market equity is $ME_t(n) = P_t(n)S_t(n)$, where $P_t(n)$ and $S_t(n)$ denote the price and shares outstanding. Each asset has K exogenous characteristics $X_t(n) \in \mathbb{R}^K$. Log market equity $me_t(n)$ is endogenous, determined in equilibrium by investor demand.

A Investor Demand

There are I institutional investors ($i = 1, ..., I$), a representative household sector (HH), and a representative Robinhood trader (RH). All $I + 2$ investors $j = \{i\}_{i=1}^{I}, HH, RH$ are short-sale constrained, have log utility and heterogeneous beliefs about expected returns and covariances, and are subject to heterogenous leverage constraints. Following KY, the optimal portfolio weights $w_t^j(n)$ are log-linear in characteristics:

$$\forall j: \quad \frac{w_t^j(n)}{w_t^j(0)} = \exp\left\{\theta_t^j + \gamma^j me_t(n) + \sum_{k=1}^{K} \beta_k^j X_{k,t}(n)\right\}\epsilon_t^j(n), \tag{4}$$

where $w_t^j(0)$ is the weight on the outside asset and θ_t^j represents the time-varying demand for stocks inside the investment universe relative to the outside asset. Equation (4) can be estimated empirically using observed portfolio holdings.

B Demand Elasticities

Let $\mathbf{p}_t, \mathbf{q}_t^j \in \mathbb{R}^N$ denote vectors of log prices and log shares held by investor j. Then the demand elasticity with respect to price is

$$-\frac{\partial \mathbf{q}_t^j}{\partial \mathbf{p}_t'} = \mathbf{I} - \gamma^j \text{diag}(\mathbf{w}_t^j)^{-1} \mathbf{G}_t^j, \tag{5}$$

with $\mathbf{G}_t^j = \text{diag}(\mathbf{w}_t^j) - \mathbf{w}_t^j \mathbf{w}_t^{j\prime}$. The parameter γ^j governs price elasticity: smaller γ^j implies greater willingness to accommodate demand shocks, that is, more liquidity provision. The set $\left\{\{\gamma^i\}_{i=1}^I, \gamma^{\text{HH}}, \gamma^{\text{RH}}\right\}$ how retail demand transmits into the cross-section of prices. If $\gamma^j = 1$ for all investors, i.e., the equity market is perfectly inelastic, then retail demand shocks have an infinite impact on the price. Lastly, note that if j holds a diversified portfolio, then $\mathbf{w}_t^j \mathbf{w}_t^{j\prime}$ is small and the diagonal elements of $-\frac{\partial \mathbf{q}_t^j}{\partial \mathbf{p}_t'}$ are approximately $-\frac{\partial q_t^j(n)}{\partial p_t(n)} = 1 - \gamma_j$.

C Market Clearing

The system of $I+2$ demand functions given by (4) is linked via the market clearing condition for each stock n, which implies that $ME_t(n) = \sum_j A_t^j w_t^j(n)$ for each n, where A_t^j is the AUM of investor j. Plugging in (4) and rewriting in vector notation, the equilibrium market clearing prices $\mathbf{p}_t$ solve

$$\mathbf{p}_t = \log\left(\sum_j A_t^j \mathbf{w}_t^j(\mathbf{p}_t)\right) - \mathbf{s}_t, \tag{6}$$

where $\mathbf{s}_t \in \mathbb{R}^N$ is the vector of log shares outstanding. The dependence of $\mathbf{w}_t^j(\mathbf{p}_t)$ on $\mathbf{p}_t$ reveals how the explicit estimation of a demand function for each investor j allows to address the interplay of demand and equilibrium prices. When changing, e.g., Robinhood traders' demand, equation (6) explicitly accounts for the endogenous demand responses by all other investors. In other words, Robinhood traders' demand affects the portfolio choice of all other investors because equilibrium prices enter their demand functions via γ^j.

3.2 Estimating the Demand System

A Instrument Construction

As in KY (2019), we treat characteristics $X_t(n)$ as exogenous to latent demand $\epsilon_t^j(n)$ but prices as endogenous. To address this, we construct investor-specific instruments for log market equity $\widehat{me}_t^j(n)$, exploiting heterogeneity in the investment universes across investors. Let $\mathcal{N}_t^j$ be investor j's investment universe. We define $\widehat{\mathcal{N}}_t^j$ as the part of investor j's universe that is exogenously determined, constructed as the set of stocks held continuously for at least eight quarters. The instrument is the counterfactual log market equity if all investors held equal-weighted portfolios over their exogenous investment universe:

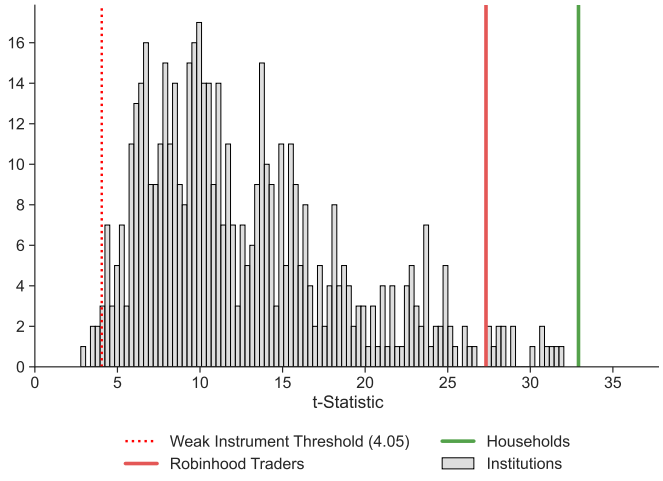
$$\widehat{me}_t^j(n) = \sum_{i \neq j,\, \mathrm{RH,HH}} A_t^i \frac{\mathbb{1}_t^i(n)}{1 + |\widehat{\mathcal{N}}_{t,i}|}. \tag{7}$$

where $\mathbb{1}_t^i(n)$ is an indicator equal to 1 if stock n is in $\widehat{\mathcal{N}}_{t,i}$. Note that the exogeneity of the instrument also relies on a wealth distribution $\{A_t^i\}_{i=1}^I$ that is orthogonal to the latent demand shocks driving portfolio weights. One could argue that latent demand, such as ESG preference shocks, potentially drive both prices and flows across institutional investors. Therefore, we construct an exogenous wealth distribution that would be obtained if assets

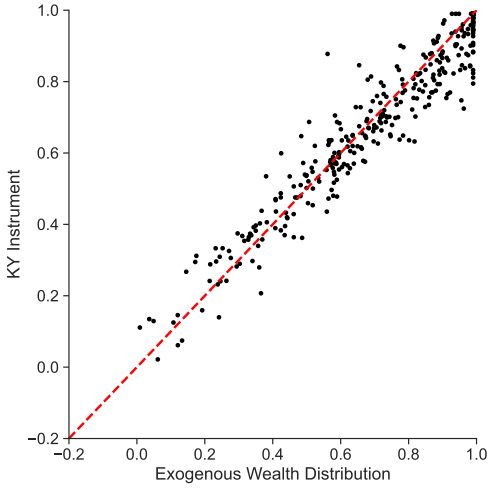
under management were proportional to the number of stocks held, $|\mathcal{N}_t^j|$.[11] We assess the instrument's relevance for each investor in a panel regression of $\widehat{me}_t^j(n)$ onto $me_t(n)$ and the exogenous characteristics $X_t(n)$ including time-fixed effects. Panel (a) of Figure 2 plots the first stage t-statistic of the instrument for institutions, households, and Robinhood traders.

Figure 2: **Weak Instrument Test**
Panel (a) plots the first stage t-statistic on the instrument $\widehat{me}_t^j(n)$ for institutions, households, and Robinhood traders. The red dotted line indicates the weak instrument threshold of 4.05 (see Stock and Yogo (2002)). Panel (b) plots the estimated γ^j for all investors under the different instrument specifications.

(a) 1st Stage t-Statistic



(b) Estimates of γ^j for different instruments



We can confidently reject the null hypothesis of weak instruments for households, Robinhood traders, and almost all institutions.[12] If we construct the instrument under this alternative wealth distribution and investment universe, the resulting estimates for γ^j remain in the same ballpark as in KY. Panel (b) plots the γ^j estimates for all investors under the

[11] Formally, the instrument then becomes

$$\widehat{me}_t^j(n) = \sum_{i \neq j, \text{RH}, \text{HH}} |\mathcal{N}_t^i| \frac{\mathbb{1}_t^i(n)}{1 + |\widehat{\mathcal{N}}_{t,i}|}$$

[12] The few institutions with marginally insignificant first-stage coefficients have a combined AUM of $10 billion and are pooled together with the household sector.

different instrument specifications. The estimates are of the same order of magnitude and strongly positively correlated ($\rho = .98$) across institutions. We conclude that our results are robust to the construction of the instrument.

B Estimation

We estimate demand in (4) by constrained generalized method of moments (GMM) under the moment condition

$$\forall j: \quad \mathrm{E}\left[\log(\epsilon_t^j(n)) \;\middle|\; \widehat{me}_t^j(n), X_t(n)\right] = 0 \quad s.t. \quad \gamma^j < 1. \tag{8}$$

where the restriction ensures downward-sloping demand and a unique equilibrium price. Following Koijen and Yogo (2019), we estimate demand over the *panel* of holdings including investor-time fixed effects. We include log book equity, profitability, investment, market beta, and dividend-to-book as exogenous characteristics. Robinhood demand is estimated from June 2018 through June 2020, and institutional and household demand over the full panel of available data from January 2000 through June 2020. Institutions with sparse portfolios are pooled by type, size, and style (see Appendix B). Taking logs on both sides of equation (4), we reformulate the demand function into a panel estimation problem for each investor j including time fixed effects.

C Estimation Results

Table 1 summarizes the estimated demand curves. For illustrative purposes, we group 13F investors—which comprise over 3,500 institutions—by their respective institutional types: Hedge funds, banks, pension funds, investment advisors (including mutual funds), insurance

companies, and other. We report the AUM-weighted average demand coefficients across investors within a group. Note that from the estimated demand coefficients γ_j, we can infer each investor's demand elasticity with respect to the price via equation (5).

Table 1: **Estimated Demand Functions**
The table reports the estimated demand coefficients for Robinhood traders, households, and 13F institutions. Robinhood demand is estimated from June 2018 through June 2020, and institutional and household demand from January 2000 through June 2020. The disaggregated nature of the 13F data allows estimating a separate demand function for each institution. We group 13F institutions into 6 groups based on Refinitiv's type labels (hedge funds, banks, insurance companies, investment advisors, pension funds) and report the AUM-weighted average for each group.

		Estimated Coefficients						
	Wealth Share	γ^j	β^j_{be}	$\beta^j_{\text{inv.}}$	β^j_{profit}	β^j_{beta}	$\beta^j_{\text{div/be}}$	Demand Elasticity
Institutional Investors								
Pension Fund	4%	0.98	16.87	-2.19	0.45	-3.11	0.05	**0.03**
Bank	6%	0.98	32.69	0.36	0.26	-5.42	0.25	**0.04**
Investment Advisor	48%	0.91	27.75	3.42	0.36	-0.03	0.10	**0.10**
Other	1%	0.86	48.67	3.04	0.64	-0.25	0.18	**0.16**
Insurance Company	2%	0.82	0.23	4.66	-1.89	-5.31	-1.01	**0.22**
Hedge Funds	2%	0.67	0.55	7.88	0.71	6.64	-0.35	**0.35**
Households	37%	0.81	3.01	0.63	0.16	-5.93	0.07	**0.19**
Robinhood Traders	0.04%	0.22	-19.42	7.06	-0.81	28.03	-0.08	**0.79**

Elasticity and Liquidity Provision Among all investor groups, Robinhood traders exhibit the highest demand elasticity. Their elasticity estimate of .79 implies that, ceteris paribus, a 10% increase in the price of stock n leads Robinhood traders to sell roughly 8% of their holdings in that stock. As we show later, their demand response with respect to prices is more nuanced than what can be captured by a single constant elasticity. The liquidity-providing role of Robinhood traders is consistent with prior evidence on retail contrarian behaviour (Kaniel et al., 2008, 2012; Kelley and Tetlock, 2013; Barrot et al., 2016). The structural demand estimation has the advantage of comparing this elasticity to other

institutional groups and quantifying its equilibrium effects. Robinhood traders are 2.25 times more elastic than hedge funds, who have an estimated elasticity of 0.35. Pension funds and banks, by contrast, are the least responsive, reflecting their long-term, largely passive mandates. Roughly 60% of the US equity market is held by investment advisors, banks, and pension funds—groups that respond only weakly to price movements.

Demand for Small and Risky Stocks The coefficients on book equity, β^j_{be}, and market beta, β^j_{beta}, capture investors' demand for size and risk, respectively. While investment advisors and banks exhibit a preference for large, high-dividend stocks, hedge funds tilt toward smaller, low-dividend stocks. Robinhood traders display an extreme preference for small-cap equities, which contributes to their outsized price impact on the smallest quintile of US stocks (see Section 4.1). Their demand coefficient for book equity is -19, more than thirty times larger in magnitude than hedge funds' corresponding estimate. Similarly, Robinhood traders exhibit an outsized demand for risky stocks, with their coefficient on market beta roughly four times larger than that of hedge funds—the investor group with the next highest preference for market risk. Demand for market risk, as captured by β^j_{beta}, is lower among liability-constrained institutions such as insurance companies and pension funds, and highest among Robinhood traders.

Robinhood Demand versus Other Households The demand functions of Robinhood traders and other households diverge sharply. For nearly all stock-specific characteristics, their estimated demand coefficients have opposite signs. This contrast may, in part, reflect the design of the trading platform and the way information is displayed on Robinhood (Barber et al., 2022a). Importantly, these diverging preferences are not specific to the lockdown period associated with COVID-19. Estimating Robinhood traders' demand curves prior to

2020 yields extremely similar coefficients. As shown in Table 2, the pre-COVID estimates closely resemble those from the full sample, suggesting that the behavioral differences predate the pandemic surge in retail trading. Overall, the portfolio of Robinhood traders—with its strong tilt toward small, risky stocks—resembles that of hedge funds more than that of traditional households. We later show that their demand differs from hedge funds along a key dimension: sensitivity to recent returns. Unlike hedge funds, Robinhood traders actively chase both recent winners and losers, consistent with attention-driven trading documented by Barber et al. (2022a). These results highlight the need for a more nuanced view of household behavior. As retail trading continues to expand and a larger share of household wealth is allocated via mobile platforms, the aggregate household sector may increasingly shift toward distinct retail habitats. The next section quantifies the equilibrium pricing implications of surging Robinhood assets and the rise of retail trading more broadly.

D Robustness to Alternative Portfolios and Time Periods

Table 2 reports the estimated demand coefficients for alternative constructions of the Robinhood portfolio $w_t^{\text{RH,prc}}(n)$ and $w_t^{\text{RH,poly}}(n)$ and $w_t^{\text{RH,exp}}(n)$. Robinhood traders' pronounced preference for small and risky stocks, as well as their high price elasticity of demand, are not an artifact of the construction of the Robinhood portfolio. Other Robinhood portfolios yield similarly extreme preferences. Across all portfolios, Robinhood traders demand for small-cap and high-beta stocks greatly exceeds that of any other investor group.

Table 2: **Robinhood Traders' Demand: Robustness**
The table reports the estimated demand coefficients for Robinhood traders using the 4 different portfolio proxies: The baseline Robinhood portfolio $w_t^{\text{RH}}(n)$, the price-weighted Robinhood portfolio $w_t^{\text{RH,prc}}(n)$, the fitted Robinhood portfolio based on a cubic fit $w_t^{\text{RH,poly}}(n)$, and the fitted Robinhood portfolio based on an exponential fit $w_t^{\text{RH,exp}}(n)$. The sample period is 2018 to 2020. The bottom row reports the baseline Robinhood demand estimated using only data before the COVID period, i.e., before 2020.

	Estimated Coefficients						Demand Elasticity
	γ^j	β^j_{be}	$\beta^j_{\text{inv.}}$	β^j_{profit}	β^j_{beta}	$\beta^j_{\text{div/be}}$	
Alternative Portfolio Proxies							
Original	0.22	-19.42	7.06	-0.81	28.03	-0.08	**0.79**
Price-weighted	0.73	-17.24	13.88	-0.53	17.31	-0.45	**0.34**
Polynomial	0.22	-12.54	3.60	-3.41	10.00	-0.89	**0.78**
Exponential	0.40	-36.94	12.03	-1.45	52.88	-0.08	**0.61**
Pre-2020 Coefficients	0.22	-18.77	8.77	-1.23	20.97	4.15	**0.78**

Notably, although these alternative portfolio proxies are far from perfectly correlated, their tilts move in similar directions, yielding broadly consistent estimated demand curves. The next section highlights that the estimated price impact of Robinhood traders remains robust across all of these alternative portfolio constructions.

The extreme coefficients are not driven by the extraordinary COVID period, during which many stock characteristics reached unusual levels. The bottom row of Table 2 reports the estimated demand coefficients using only pre-2020 data. The resulting coefficients are largely unchanged, suggesting that the pronounced Robinhood demand for small-cap and risky stocks is not specific to the COVID episode but rather a general feature of (gamified) retail trading. Appendix Table D.1 reports the estimated institutional coefficients for the pre-2020 period, which are also remarkably stable over time.

4 The Impact of Robinhood Traders on the Equity Market

Having estimated each investor's demand curve, we can express equilibrium prices via the market clearing condition (6) as a function of supply and demand components. The market clearing condition is a fixed point problem and allows computing the counterfactual equilibrium prices that would clear equity markets under alternative inputs.[13] For example, we can assess the counterfactual equilibrium obtained without the presence of Robinhood traders by setting $A_t^{\text{RH}} = 0$ and redistributing their wealth across all other investors. The counterfactual equilibrium price vector $\mathbf{p}_t \in \mathbb{R}^N$ is an implicit function $\mathbf{g}(\cdot)$ of exogenous characteristics and shares outstanding, as well as the AUM and demand shocks of all investors,

$$\mathbf{p}_t = \mathbf{g}\Bigg(\underbrace{\mathbf{X}_t, \mathbf{s}_t}_{\text{Supply}}, \underbrace{\{A_t^j, \boldsymbol{\epsilon}_t^j\}_{j=\text{I,HH,RH}}}_{\text{Demand}} \Bigg). \tag{9}$$

The following sections use equation (9) in three applications to quantify the marginal and total price impact of Robinhood traders, both generally and during specific episodes such as during the stock market rally in the second quarter of 2020.

4.1 Robinhood Traders Impact on Equity Prices

The equilibrium pricing function (9) allows us to infer the marginal and absolute impact of different investors on equity prices. To this end, note that the market clearing condition allows us to extract the equilibrium price vector $\mathbf{p}_t^{(i)}$ that clears equity markets at time t without investor i by redistributing their assets across all other investors. This is similar to assuming the investor's assets were instead invested in the market portfolio. However,

[13] We numerically solve for counterfactual equilibria using the algorithm proposed in KY (2019). Uniqueness is guaranteed by the coefficient constraint that $\gamma^j < 1$ for all investors.

because market prices are endogenously determined in equilibrium, the correct approach is to redistribute assets according to the wealth share of other investors. For example, $\mathbf{p}_t^{(\text{RH})}$ is the equilibrium price vector that clears equity markets without Robinhood traders. This assumes that without the existence of the brokerage platform, the shares traded on Robinhood would be held by all other investors. In the next section we consider alternative counterfactuals. Formally, the counterfactual equilibrium price vector without investor type i is given by

$$\mathbf{p}_t^{(i)} = \mathbf{g}\left(\mathbf{X}_t, \mathbf{s}_t, \{\gamma^j, \beta^j, \tilde{A}_t^j, \boldsymbol{\epsilon}_t^j\}_{j \neq i}\right), \tag{10}$$

where $\tilde{A}_t^j = A_t^j + A_t^i \frac{A_t^j}{\sum_{k \neq i} A_t^k}$ is the counterfactual wealth distribution in the absence of investor i. Using the counterfactual price vector $\mathbf{p}_t^{(i)}$, we can gauge the *marginal* contribution of each investor as the total repricing relative to the investor's wealth share. Following Koijen et al. (2024), we define total equity repricing due to investor i as

$$\text{Repricing}_t^{(i)} = \frac{\sum_{n=1}^N |ME_t^{(i)}(n) - ME_t(n)|}{\sum_{n=1}^N ME_t(n)} \tag{11}$$

where $ME_t^{(i)}(n) = \exp\{p_t^{(i)}(n) + s_t(n)\}$ is the counterfactual market equity in the absence of investor i. This is equivalent to the value-weighted absolute percentage change in prices. Dividing by the investor's wealth share yields the marginal impact of an additional dollar invested with that investor, i.e., the repricing per dollar wealth of i. Formally, the marginal impact is given by

$$\text{Marginal Impact}_t^{(i)} = \frac{\text{Repricing}_t^{(i)}}{\text{Wealth Share}_t^i} \tag{12}$$

where an investor's wealth share is given by their assets relative to the total assets, $\frac{A_t^i}{\sum_{i=1}^I A_t^i}$.

Table 3: **Total Repricing and Marginal Impact of Different Investor Types**
The first column reports the wealth share of each investor group. The second column reports the repricing of each investor type, given by the value-weighted absolute percent change in prices when their assets are redistributed across other investor types. The third column reports the marginal price impact given by the repricing divided by the wealth share. Panel (a) reports the values for all stocks. Panel (b) splits the sample into quintiles by market capitalization. The corresponding wealth share is computed as the total dollar holdings by Robinhood Traders in stocks in each respective quintile relative to the aggregate market capitalization of stocks in that quintile. Values are the time series average across quarters over the sample from 2018 through 2020.

Panel (a) All Investors' Repricing

Investor Type	Wealth Share (%)	Repricing (%)	Marginal Impact *(Repricing per dollar wealth)*
Households	36.83	100.74	2.74
Investment Advisors	48.06	93.92	1.95
Banks	6.38	13.91	2.18
Hedge Funds	2.37	10.70	4.51
Insurance Companies	1.56	10.29	6.61
Pension Funds	3.78	6.02	1.59
Other	0.99	3.14	3.18
Robinhood Traders	0.04	0.25	6.74

Panel (b) Robinhood Traders' Repricing by Size Quintile

Size Quintile	Wealth Share (%)	Repricing (%)	Marginal Impact *(Repricing per dollar wealth)*
Quintile 1 (Small)	2.23	7.66	3.44
Quintile 2	0.57	2.50	4.34
Quintile 3	0.18	0.90	5.14
Quintile 4	0.05	0.38	7.39
Quintile 5 (Large)	0.02	0.20	9.72

Despite estimating over different sample periods, the marginal impact (repricing per dollar of wealth) for institutional investors closely aligns with the estimates in Koijen et al. (2024) (Table 6).[14] We estimate a repricing ratio of 4.5 for hedge funds, which implies that each dollar invested with hedge funds leads to $4.5 of total equity repricing. Quite strikingly, Robinhood traders exhibit the highest marginal impact of all investor groups:

[14]For example, our estimates for hedge funds of $4.5 are around twice that of banks at $2.2, compared to $3.6 and $1.81, respectively, in Koijen et al. (2024).

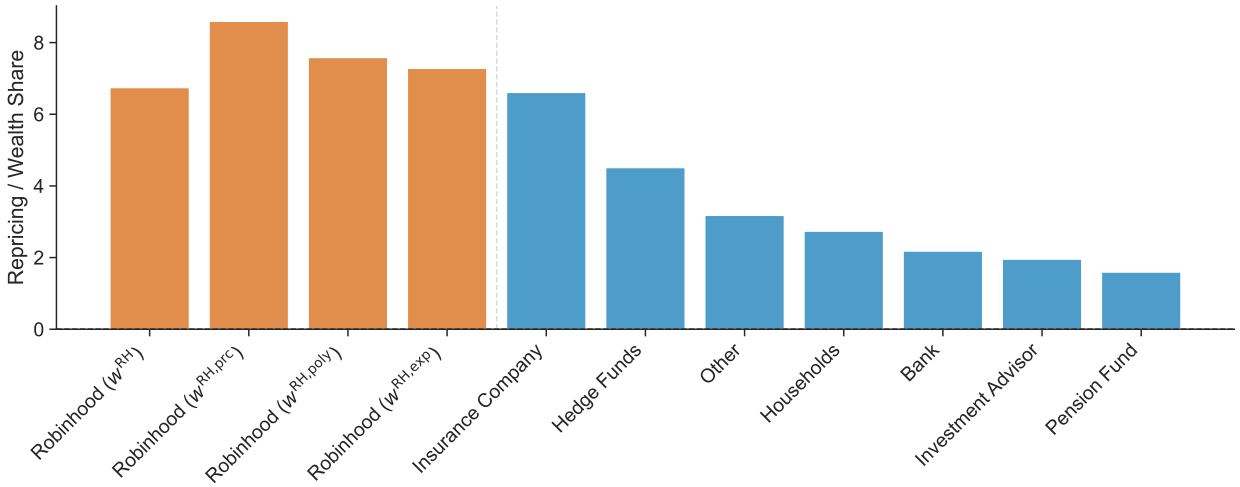
each dollar invested by Robinhood induces \$6.74 of total equity repricing, exceeding hedge funds (\$4.51) and insurance companies (\$6.61) as well as all other institutional categories and households. This high marginal impact reflects how far their demand curve deviates from the asset-weighted average of all other investors. Their extreme portfolio tilts translate into considerable repricing per dollar of wealth. However, Robinhood traders' total market share amounts to only 0.04% of total equity wealth, so their aggregate repricing effect in value-weighted terms remains small. Nevertheless, their large marginal impact suggests that continued growth in retail trading activity could have increasingly material consequences for the cross-section of equity prices.

Panel (b) of Table 3 confirms that Robinhood traders' *total* repricing is concentrated in small-cap stocks, where their ownership share is considerably higher. Redistributing Robinhood traders' assets leads to a total repricing of small-cap stocks of 7.66%. The impact declines monotonically moving to larger size quintiles. Interestingly, however, their *marginal* impact exhibits the opposite pattern: it increases monotonically with size (from 3.44 in the bottom quintile to 9.72 in the top quintile). This increase reflects that larger stocks are disproportionately held by passive investors; a marginal dollar of Robinhood flow therefore confronts a steeper residual demand curve in large stocks, producing larger per-dollar price effects.

Figure 3 illustrates the marginal impact of Robinhood traders relative to other investor groups and confirms that the repricing effect is not an artifact of the specific construction of the Robinhood portfolio. Although the four alternative portfolio constructions are only moderately correlated, each yields a marginal impact that is substantially higher than that of any other institutional investor group or households. The pronounced marginal impact of Robinhood traders—driven by their distinct preferences for stock characteristics and their

higher price elasticity—emerges consistently across all portfolio constructions.

Figure 3: **Marginal Impact of Robinhood Traders**
The figure reports the marginal impact of Robinhood traders relative to other investor types. Marginal impact is defined as the total repricing per dollar of wealth. The four bars on the left report the total marginal impact of Robinhood traders under all four portfolio constructions: The base portfolio w^{RH}; the price-weighted portfolio, $w^{RH,prc}$; the cubic adjusted $w^{RH,poly}$; and the exponential fit $w^{RH,exp}$. The remaining bars plot the marginal impact of other investor types computed by redistributing their assets proportionately across all remaining investor types.



4.2 Repricing under Alternative Counterfactuals

The previous section quantified the marginal repricing of equities that would occur if Robinhood traders' assets were redistributed across all other investors. We now consider three alternative counterfactuals. First, we redistribute Robinhood assets to other households, approximating a scenario in which, absent the surge in Robinhood trading, these assets would have been held through traditional brokerages rather than mobile trading platforms. Second, we redistribute assets to investment advisors, implying that Robinhood traders would instead have invested their wealth through financial intermediaries such as 401(k) plans. Finally, we consider a scenario in which, without Robinhood, these funds would not have entered the

equity market at all but instead remained in deposits or were consumed.[15]

Table 4 reports the resulting repricing for each counterfactual and compares it to the baseline case in which assets are redistributed with respect to the market portfolio. The overall magnitudes are broadly similar across the different counterfactuals: a marginal dollar invested via Robinhood leads to an aggregate equity repricing of between \$6 and \$9, well exceeding that of any other investor group. This large impact arises because Robinhood traders' portfolios deviate sharply from the asset-weighted demand curve of all other investors. Consequently, regardless of which investor group the Robinhood assets are redistributed to, the resulting repricing remains substantial.

Table 4: **Robinhood Traders' Marginal Impact: Alternative Counterfactuals**
The table reports the marginal impact (repricing per dollar wealth) of Robinhood traders under different portfolio constructions and counterfactual experiments. The first row is a counterfactual where Robinhood traders' AUM is proportionally redistributed across other investor types, as in Table 3. The second and third rows denote counterfactuals where Robinhood traders' AUM is distributed only to households or to investment advisors, respectively. The final row denotes a counterfactual where Robinhood traders' AUM is withdrawn from the market. The first column reports results for the standard Robinhood portfolio, w^{RH}. The second column reports results for the price-weighted portfolio, $w^{\text{RH,prc}}$. The third and fourth columns report results for the cubic and exponentially adjusted portfolios, $w^{\text{RH,poly}}$ and $w^{\text{RH,exp}}$, respectively.

Counterfactual	w^{RH}	$w^{\text{RH,prc}}$	$w^{\text{RH,poly}}$	$w^{\text{RH,exp}}$
Proportionally redistribute RH AUM	6.74	8.59	7.58	7.28
Distribute RH AUM to households	6.87	8.41	7.62	7.37
Distribute RH AUM to Investment Advisors	6.75	8.77	7.60	7.28
Withdraw RH AUM	5.96	7.92	6.76	6.49

4.3 Robinhood Traders' Impact on Volatility

A large empirical literature on the interaction of stock returns and retail trading activity documents short-term contrarian behaviour by individual investors (see e.g. Kaniel et al. (2008) and Barber et al. (2008)). More recently, Peress and Schmidt (2020) find that transitory

[15]Entirely withdrawing Robinhood traders' AUM leads to a mechanical (but small) decline in aggregate valuations, while under redistribution the value of the aggregate equity market remains unchanged.

declines in the intensity of noise trading lead to deteriorating liquidity and higher volatility.[16] In this section, we use the structural model to quantify the impact of retail trading on the volatility of stock returns, which would be difficult to do in a reduced-form framework.

The previous section highlighted that Robinhood traders have the largest *marginal* impact of all investor groups. Their total impact on the volatility of stock returns depends on how their demand *changes* over time, as well as their total assets. To quantify Robinhood traders' contribution to market volatility, we focus on the second quarter of 2020—a period marked by an extraordinary surge in retail trading and record inflows to mobile-first retail brokerages. This episode provides a natural laboratory for assessing how rapid shifts in retail demand translate into return volatility.

We first use equation (9) to decompose the cross-sectional volatility of stock returns into supply- and demand-driven components. We label changes in the exogenous characteristics $\mathbf{X}_t$ and log shares outstanding $\mathbf{s}_t$ as *supply* changes. *Demand* changes refer to changes in assets under management $(A_t^{\mathrm{I}}, A_t^{\mathrm{HH}}, A_t^{\mathrm{RH}})$ and changes in latent demand $(\boldsymbol{\epsilon}_t^{\mathrm{I}}, \boldsymbol{\epsilon}_t^{\mathrm{HH}}, \boldsymbol{\epsilon}_t^{\mathrm{RH}})$. To reduce the notational clutter, we define the set of supply and demand variables as $\mathbf{S}_t = (\mathbf{X}_t, \mathbf{s}_t)$ and $\mathbf{D}_t^j = (A_t^j, \boldsymbol{\epsilon}_t^j)$ respectively. By iteratively changing supply and investor-specific demand to their next period's values, we can decompose the vector of log returns $\mathbf{r}_{t+1} = \mathbf{p}_{t+1} - \mathbf{p}_t$ into

[16] Using a French reform that permanently discouraged retail trading for a subset of stocks, Foucault et al. (2011) find that reduced retail trading activity improves liquidity and decreases volatility. Similarly, Eaton et al. (2022) find that intense herding by retail traders may lower market quality.

its subcomponents,

$$
\begin{aligned}
\mathbf{r}^{\mathrm{S}}_{t+1} &= \mathbf{g}\Big(\mathbf{S}_{t+1}, \mathbf{D}^{\mathrm{I}}_{t}, \mathbf{D}^{\mathrm{HH}}_{t}, \mathbf{D}^{\mathrm{RH}}_{t}\Big) - \mathbf{p}_{t},\\
\mathbf{r}^{\mathrm{I}}_{t+1} &= \mathbf{g}\Big(\mathbf{S}_{t+1}, \mathbf{D}^{\mathrm{I}}_{t+1}, \mathbf{D}^{\mathrm{HH}}_{t}, \mathbf{D}^{\mathrm{RH}}_{t}\Big) - \mathbf{g}\Big(\mathbf{S}_{t+1}, \mathbf{D}^{\mathrm{I}}_{t}, \mathbf{D}^{\mathrm{HH}}_{t}, \mathbf{D}^{\mathrm{RH}}_{t}\Big),\\
\mathbf{r}^{\mathrm{HH}}_{t+1} &= \mathbf{g}\Big(\mathbf{S}_{t+1}, \mathbf{D}^{\mathrm{I}}_{t+1}, \mathbf{D}^{\mathrm{HH}}_{t+1}, \mathbf{D}^{\mathrm{RH}}_{t}\Big) - \mathbf{g}\Big(\mathbf{S}_{t+1}, \mathbf{D}^{\mathrm{I}}_{t+1}, \mathbf{D}^{\mathrm{HH}}_{t}, \mathbf{D}^{\mathrm{RH}}_{t}\Big),\\
\mathbf{r}^{\mathrm{RH}}_{t+1} &= \mathbf{p}_{t+1} - \mathbf{g}\Big(\mathbf{S}_{t+1}, \mathbf{D}^{\mathrm{I}}_{t+1}, \mathbf{D}^{\mathrm{HH}}_{t+1}, \mathbf{D}^{\mathrm{RH}}_{t}\Big).
\end{aligned}
$$

Note that $\mathbf{g}(\cdot)$ is concave in investors' dollar demand. Thus we purposely place the change in Robinhood traders' demand *after* having accounted for institutions' and households' demand change in order to leave the least scope for Robinhood traders to affect volatility. Using the fact that $\mathbf{r}_{t+1} = \mathbf{r}^{\mathrm{S}}_{t+1} + \mathbf{r}^{\mathrm{I}}_{t+1} + \mathbf{r}^{\mathrm{HH}}_{t+1} + \mathbf{r}^{\mathrm{RH}}_{t+1}$, we decompose the volatility of stock returns during the second quarter of 2020 into its underlying supply and demand components,[17]

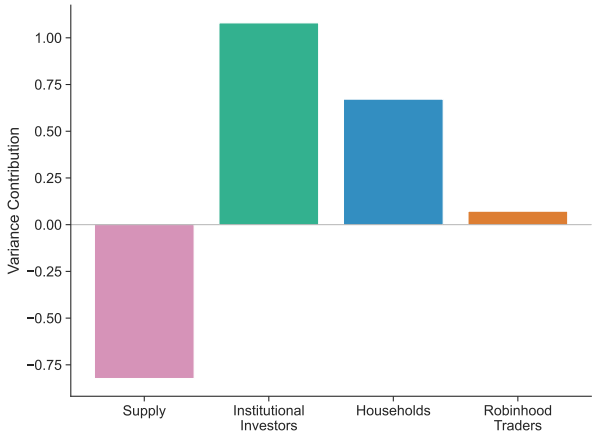
$$
\mathrm{Var}(\mathbf{r}_{t+1}) = \mathrm{Cov}(\mathbf{r}_{t+1}, \mathbf{r}^{\mathrm{S}}_{t+1}) + \mathrm{Cov}(\mathbf{r}_{t+1}, \mathbf{r}^{\mathrm{I}}_{t+1}) + \mathrm{Cov}(\mathbf{r}_{t+1}, \mathbf{r}^{\mathrm{HH}}_{t+1}) + \mathrm{Cov}(\mathbf{r}_{t+1}, \mathbf{r}^{\mathrm{RH}}_{t+1}). \tag{13}
$$

The structural nature of the decomposition explicitly takes into account the endogeneity of prices and demand, and therefore allows for a quantification of Robinhood traders' contribution to volatility that goes beyond simple regressions of returns onto holdings as in e.g. Ozik et al. (2021) or Glossner et al. (2025). Figure 4 reports the decomposition of the cross-sectional variation into changes in supply (i.e. changes in exogenous characteristics and shares outstanding) and demand (i.e. changes in the AUM and latent demand of institutions, households, and Robinhood traders).

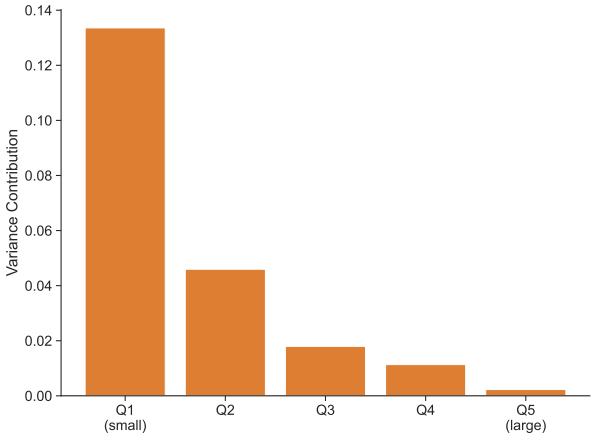
[17] The variance and covariance terms are purely *cross-sectional*, i.e. they are computed as $\mathrm{Var}(\mathbf{r}_{t+1}) = \frac{1}{N}\sum_{n=1}^{N}\big(r_{t+1}(n) - \bar{r}_{t+1}\big)^2$.

Figure 4: **Robinhood Traders' Impact on Volatility in 2020**
This figure reports the decomposition of variance during the second quarter of 2020. Variance is decomposed into investor-specific subcomponents, which add up to 1 as given by equation (13). The *Supply* bar indicates the variation in returns explained by changes in fundamentals and shares outstanding. The *Demand* bars (split into institutions, households, and Robinhood traders) indicate the variation explained by changes in latent demand and assets under management. Panel (a) reports the variance decomposition across all stocks. Panel (b) reports the contribution of Robinhood traders by market capitalization quintile.

(a) Variance Decomposition Across All Stocks



(b) Robinhood Contribution by Size Quintile



Panel a) reports the variance decomposition across all stocks. The impact of Robinhood traders on volatility during the second quarter of 2020 is substantial. Their shifts in latent demand account for over 7% of the aggregate volatility in the second quarter of 2020. This emphasizes the impact of large swings in Robinhood traders' demand curve beyond the stable coefficient on fundamental characteristics documented in Section 3.2.C. However, their impact is largely concentrated in small firms. Panel b) plots the splits the contribution of Robinhood traders to the volatility by size quintile. While Robinhood traders drove over 13% of the volatility of the the smallest stocks (bottom quintile) their impact on large caps (the top quintile) is negligible. In other words, the volatility of small cap stocks would have been over 13% lower without the additional demand from Robinhood traders, whereas the volatility of large caps would have remained largely unchanged. The incremental contribution of

Robinhood traders to volatility decreases monotonically with size. This is owed to Robinhood traders' reduced ability to affect the valuation of large cap stocks, which account for the majority of the aggregate market capitalization, due to their limited wealth share in these stocks. Even with an extremely inelastic institutional sector, the retail sector's dollar demand shocks are too small relative to the market capitalization of large cap stocks.

5 Attention-Induced Latent Demand

Motivated by the importance of latent demand in driving the volatility of small-cap stocks, we now turn to the determinants of latent demand for Robinhood traders. Barber and Odean (2008) show that retail traders have a preference for both extreme winners and losers. Building on this evidence, Barber et al. (2022b) highlight that Robinhood traders engage in more attention-induced trading than other retail investors. They show that Robinhood traders tend to herd in the "Top Mover" list displayed on the mobile application, which lists the stocks with the largest absolute percentage price changes from the previous day close. Motivated by their findings, we examine to what extent latent demand by Robinhood traders can be explained by attention-induced trading.

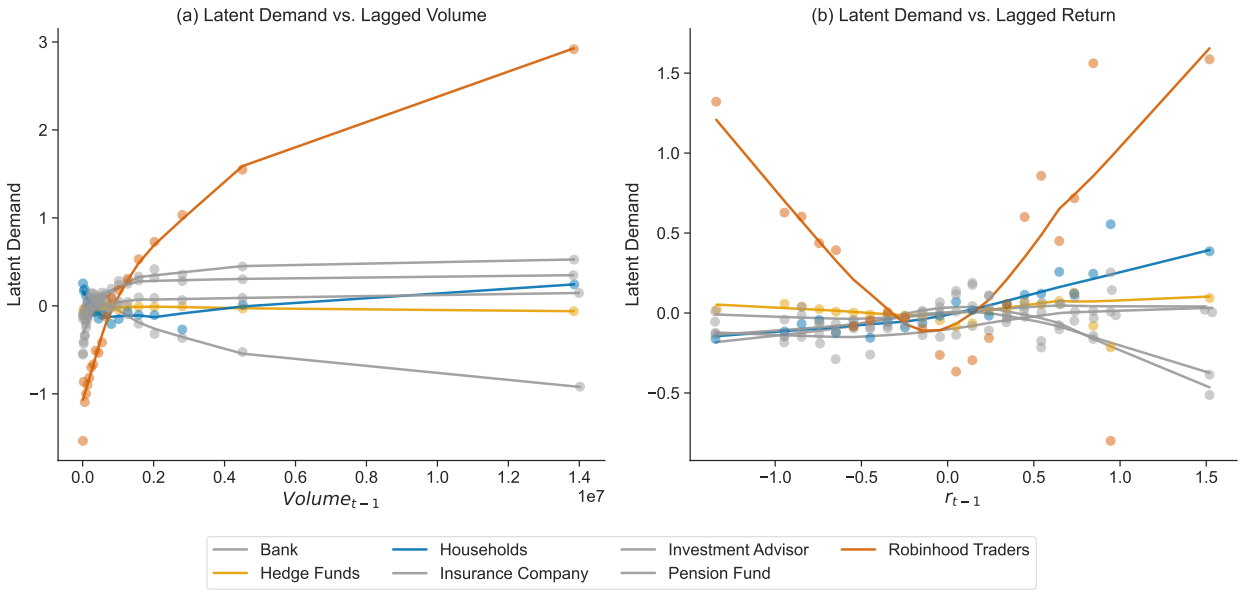
5.1 Drivers of Latent Demand

We focus on two simple proxies for investor attention: past absolute returns and past trading volume. Both capture the visibility of a stock among retail investors, as large price movements and unusually high turnover tend to draw media coverage, platform visibility, and heightened retail interest.[18]

[18]Welch (2022) similarly finds that a combination of share volume and dollar trading volume explains a significant fraction of the aggregate Robinhood portfolio weight.

Figure 5 presents a binscatter of Robinhood traders' average latent demand against past trading volume and the past absolute returns of stocks. For both trading volume and past absolute returns, a striking pattern emerges. First, Robinhood traders latent demand increases monotonically in the past trading volume across stocks. In contrast, no other institutional investor group nor households exhibits a similar pattern. Second, Robinhood traders' latent demand follows a distinct V-shaped relationship with respect to past returns: Their latent demand is high for stocks with both large negative and large positive past returns, and low for stocks with small absolute returns. In sharp contrast, the latent demand of institutional investors and other households shows little to no sensitivity to recent absolute returns.

Figure 5: **Attention-Induced Latent Demand**
Panel (a) is a binscatter of latent demand by past trading volume quantiles. Panel (b) is a binscatter of latent demand by past log return quantiles over the cross-section of all stocks in our sample. Lines are fit with locally weighted regression (LOESS).



To quantify this relationship, we regress latent demand on lagged trading volume and

lagged absolute returns for all investor groups. Table 5 reports the OLS estimates. Robinhood traders' latent demand is strongly driven by past absolute returns and past trading volume, with t-statistics of 5 and 20 respectively. These two components alone explain more than 25% of the cross-sectional variation in latent demand for Robinhood traders. In contrast, other institutional investors exhibit economically small coefficients on both past absolute returns and past trading volume. Moreover, their response to past absolute returns is reversed: their latent demand is higher for stocks that experienced lower past absolute returns.

Table 5: **Attention-Induced Latent Demand**
The table reports panel regressions including investor and time fixed effects of latent demand $\varepsilon_t^i(n)$ onto quarterly lagged absolute log returns $|r_{t-1}|$ and one-year lagged share trading volume. Each column corresponds to a different investor type: Households (HH), Banks, hedge funds (HF), insurance companies (IC), investment advisors (IA), other institutions, and pension funds (PF). t-statistics, based on standard errors double-clustered by date and institution, are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels respectively.

	Robinhood	HH	Bank	HF	Insurance Company	Investment Advisor	Other	Pension Fund
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\|r_{t-1}\|$	1.270**	-0.167***	-0.452	0.040	-0.330	-0.315**	-0.290	-0.056
	(3.54)	(-5.35)	(-1.81)	(0.39)	(-1.82)	(-3.00)	(-1.90)	(-1.01)
Volume_{t-1} (millions)	0.194***	0.013***	0.012	0.015**	0.004	0.014***	0.043***	0.014***
	(19.21)	(20.22)	(2.04)	(4.15)	(0.43)	(7.14)	(4.81)	(5.24)
Time Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Investor Fixed Effects	–	–	Y	Y	Y	Y	Y	Y
R^2 Within	0.251	0.004	0.003	0.002	0.001	0.003	0.019	0.004
Observations	9,810	9,816	266,409	174,964	32,795	1,968,685	57,917	186,613

Appendix Table D.2 confirms that this pattern is not driven by the specific construction of the Robinhood portfolio. The economically large chasing of past absolute returns and past trading volume holds robustly across all of our four proxies for the Robinhood portfolio.

This contrast highlights a distinctive feature of the recent rise in retail investing and mobile trading. Attention appears to be a central driver of Robinhood users' trading decisions, with material consequences for the volatility of small-cap stocks. Importantly, the strong

preference for large return moves is not necessarily destabilizing. When returns are positive, their trading behavior tends to amplify existing price movements. In contrast, when returns are negative, it mitigates downward pressures. For instance, during the onset of the COVID-19 pandemic in the first quarter of 2020, when many institutional investors engaged in fire sales and stock prices plummeted, Robinhood traders acted as contrarian buyers and provided considerable liquidity. The next section examines Robinhood traders' price impact at the stock level during this period, illustrating how attention-induced demand can both amplify volatility during market rallies and provide liquidity during crashes.

5.2 Stock-level Price Impact during 2020

A widespread narrative during 2020 was that some of the outsized positive returns were owed to retail investors' concentrated demand shocks. Our results present an important test of this narrative: Since Robinhood traders respond to recent volatility, i.e., both large positive *and* negative returns, we should expect to see Robinhood traders reduce large negative returns during downturns while magnifying shocks during market rallies. To this end, we use equation (10) to assess each stock's counterfactual return without the demand shocks from Robinhood traders.[19] Table 6 presents the 10 stocks in the top decile of US firms during the first half of 2020 which had the largest absolute returns and in which Robinhood traders held at least .1% of market equity.

[19]It is important for this test that we not simply remove Robinhood's AUM, which would mechanically lead to this directional result. Instead, as in Figure 3, we redistribute Robinhood's AUM proportionately to remaining investors.

Table 6: **Robinhood Traders' Price Impact at the Stock Level**
Within the largest decile of US stocks, we choose the 10 stocks with the largest absolute returns in Q1 and Q2 2020 and where Robinhood traders held at least .1% of market equity. We report the actually observed return R_{t+1} and the counterfactually observed return without Robinhood demand $R_{t+1}^{(\mathrm{RH})}(n) = \exp\{p_{t+1}^{(\mathrm{RH})}(n) - p_t(n)\} - 1$, as well as their difference ΔR_t. The last two columns in each panel decompose the price impact into the multiplier $\mathcal{M}_t(n) = 1/(1 - \sum_j z_t^j(n)\gamma^j)$ and the fraction of shares held by Robinhood traders $z_t^{\mathrm{RH}}(n)$. $R_t, R_t^{-\mathrm{RH}}(n)$, and $z_t^{\mathrm{RH}}(n)$ are reported in %.

(a) Q1

Company Name	R_{Q1}	$R_{Q1}^{\text{-RH}}$	ΔR_{Q1}	$\mathcal{M}_{Q1}(n)$	$z_{Q1}^{\mathrm{RH}}(n)$
Royal Caribbean Cruises Ltd	-75.90	-78.32	2.42	7.78	1.42
Carnival Corp	-74.09	-76.72	2.63	7.79	1.46
Halliburton Co	-72.01	-73.09	1.09	6.37	0.68
Occidental Petroleum Corp	-71.90	-72.26	0.36	5.60	0.28
MGM Resorts International	-64.53	-66.17	1.63	7.69	0.73
United Airlines Holdings Inc	-64.18	-66.02	1.84	6.79	0.92
Boeing Co	-54.22	-54.48	0.26	9.45	0.14
Delta Air Lines Inc	-51.21	-53.44	2.23	7.20	0.74
Ford Motor Co	-48.06	-56.82	8.75	8.09	2.35
General Motors Co	-43.22	-44.10	0.87	8.80	0.22

(b) Q2

Company Name	R_{Q2}	$R_{Q2}^{\text{-RH}}$	ΔR_{Q2}	$\mathcal{M}_{Q2}(n)$	$z_{Q2}^{\mathrm{RH}}(n)$
Tesla Inc	106.07	105.14	0.94	7.16	0.13
Halliburton Co	89.49	81.91	7.58	6.79	0.70
Marathon Petroleum Corp	58.26	57.29	0.97	8.76	0.14
Occidental Petroleum Corp	58.03	55.43	2.60	5.57	0.39
Royal Caribbean Cruises Ltd	56.36	34.70	21.65	8.36	1.88
ONEOK Inc	52.32	51.93	0.38	8.26	0.10
MGM Resorts International	42.37	24.04	18.33	7.84	2.02
Ford Motor Co	25.88	-1.55	27.43	8.09	3.17
Carnival Corp	24.68	-5.41	30.09	8.59	3.47
Boeing Co	22.90	21.65	1.25	8.45	0.27

Panels (a) and (b) confirm the proposed liquidity and amplification channels of attention-induced Robinhood demand. Robinhood demand substantially alleviated the negative returns observed in Q1. Ford's return during Q1 would have been nearly 9 percentage points (pp) lower without the attention-induced demand from Robinhood traders. Similarly, Carnival's share price would have dropped by an additional 2.6 pp without the demand from Robinhood. The return implications are even more pronounced during the recovery in Q2. Carnival's

return would have been 30 pp lower, and Ford's return would have been 27 pp lower without the additional demand from Robinhood traders. The magnitude of Robinhood demand effects is particularly remarkable given the size of these firms.

In order to better understand the source of the demand effects, we decompose Robinhood traders' price impact into an elasticity and ownership channel. Differentiating (6) with respect to the latent demand of Robinhood traders reveals the coliquidity matrix $\frac{\partial \mathbf{p}_t}{\partial \log(\boldsymbol{\epsilon}_t^{\mathrm{RH}})'} \in \mathbb{R}^{N \times N}$ as in KY (2019, equation 23). Assuming that $\mathrm{diag}(\mathbf{w}_j) - \mathbf{w}_j \mathbf{w}_j' \approx \mathrm{diag}(\mathbf{w}_j)$, the diagonal elements of the coliquidity matrix simplify to

$$\frac{\partial p_t(n)}{\partial \log(\epsilon_t^{\mathrm{RH}}(n))} = \mathcal{M}_t(n) z_t^{\mathrm{RH}}(n), \tag{14}$$

where $\mathcal{M}_t(n) = \frac{1}{1 - \sum_j z_t^j(n) \gamma^j}$ and $z_t^j(n)$ is the fraction of shares outstanding of n held by j (see Appendix C.1 for a proof). $\mathcal{M}_t(n)$ can be interpreted as a cross-sectional multiplier, which is inversely related to the market's price elasticity of demand for n. The more elastic the investors in stock n are, the better they accommodate Robinhood traders' demand shocks and the lower $\mathcal{M}_t(n)$. Equation (14) highlights the sources of Robinhood traders' price impact on stock n: Either the investors in n are inelastic ($\mathcal{M}_t$ is large), or Robinhood traders hold a substantial fraction of n's shares outstanding (z_t^{RH} is large), or both. The size of the cross-sectional multiplier $\mathcal{M}_t$ helps explain the large price impact of Robinhood demand despite their small ownership share z_t^{RH}.

In order to bolster intuition about the relative magnitude of the two effects, we double-sort stocks into quintiles based on ownership $z_{Q2}^{\mathrm{RH}}(n)$ and multiplier $\mathcal{M}_{Q2}(n)$. We compute the stock-specific repricing $\Delta_{Q2}^{\mathrm{RH}}(n) = \frac{M_{Q2}(n) - M_{Q2}^{-\mathrm{RH}}(n)}{M_{Q2}(n)}$ for each stock and take averages across stocks within each of the 5×5 Multiplier-Ownership portfolios. Table 7 shows the average

repricing for all portfolios in percent. In line with the decomposition of Robinhood traders'

Table 7: **Multiplier-Ownership Double Sorts**
The table reports the average repricing (%) for 5×5 multiplier-ownership quintile portfolios. As of July 2020, we sort all stocks into Robinhood ownership z_{Q2}^{RH} and multiplier $\mathcal{M}_{Q2}$ quintiles. Within each of the 25 portfolios, we compute the average repricing across the N_q stocks in portfolio $q = 1, ..., 25$ as $\frac{1}{N_q} \sum_{n \in q} \Delta_{Q2}^{\text{RH}}(n)$, where $\Delta_{Q2}^{\text{RH}}(n) = (M_{Q2}(n) - M_{Q2}^{-\text{RH}}(n))/M_{Q2}(n)$ is the repricing of stock n due to Robinhood traders.

Multiplier Quintiles $\mathcal{M}_{Q2}$	**Ownership Quintiles** z_{Q2}^{RH}				
	Low	2	3	4	High
Small	-0.27	-0.31	-0.13	1.19	22.93
2	-0.38	-0.34	-0.11	1.82	36.36
3	-0.37	-0.36	-0.13	1.94	33.12
4	-0.39	-0.37	-0.11	1.90	33.01
Large	-0.39	-0.38	-0.02	2.87	38.60

price impact (14), the average repricing conditional on ownership rises from small to big multiplier deciles. Similarly, conditional on the multiplier, the repricing increases with higher ownership. As the overall market share of Robinhood traders is small, these effects are sizeable only for the larger ownership deciles. The repricing for the portfolio comprised of the highest Robinhood-ownership stocks that are held by the most inelastic institutions is 38.6%. This implies that without the presence of Robinhood traders, the average stock in this portfolio would have had a 38.6% lower market capitalization in Q2.

6 Conclusion

This paper quantifies the impact of Robinhood traders on the US equity market within a structural demand-system framework. By estimating investor-specific demand curves, we identify Robinhood traders as a highly distinct segment of the equity market characterized by unusually elastic and attention-driven demand. Their portfolios tilt heavily toward small,

volatile stocks, diverging sharply from the preferences of both institutional investors and other households. These differences allow Robinhood traders to exert a measurable influence on equilibrium prices despite their limited aggregate wealth share.

Per dollar of invested wealth, we estimate Robinhood traders to have the largest marginal price impact of any investor group. Their high elasticity and concentrated exposure to smaller, high-volatility stocks amplifies their repricing effect in markets dominated by passive and inelastic institutions. During the first half of 2020, the surge in Robinhood assets materially contributed to variation in small-cap stock returns—accounting for roughly 13 percent of the observed cross-sectional volatility. These results highlight that even modest shifts in retail participation can meaningfully alter equity valuations and volatility.

Our results suggest that the growth of Robinhood represents a structural shift in the composition of equity demand. Even at a small fraction of the aggregate equity market, the equilibrium consequences of these investors participation are significant given the distinct nature of their demand. While our applications focused on the early surge of Robinhood's demand from 2018-2020, Robinhood's assets have grown nearly tenfold since this period, likely magnifying their impacts even further. As Robinhood and similar retail trading platforms continue to expand, their role in price formation, liquidity provision, and aggregate volatility becomes increasingly consequential.

References

Ardia, D., C. Aymard, and T. Cenesizoglu (2025). Examining high-frequency patterns in Robinhood users' trading behavior. *International Review of Financial Analysis 105*, 104369.

Asness, C. S. (2024). The less-efficient market hypothesis. *Forthcoming in the 50th Anniversary*

Issue of The Journal of Portfolio Management.

Balasubramaniam, V., J. Y. Campbell, T. Ramadorai, and B. Ranish (2023). Who owns what? a factor model for direct stockholding. *The Journal of Finance 78*(3), 1545–1591.

Barber, B. M., X. Huang, T. Odean, and C. Schwarz (2022a). Attention-induced trading and returns: Evidence from robinhood users. *The Journal of Finance 77*(6), 3141–3190.

Barber, B. M., X. Huang, T. Odean, and C. Schwarz (2022b). Attention-Induced Trading and Returns: Evidence from Robinhood Users. *The Journal of Finance 77*(6), 3141–3190.

Barber, B. M., S. Lin, and T. Odean (2024). Resolving a Paradox: Retail Trades Positively Predict Returns but Are Not Profitable. *Journal of Financial and Quantitative Analysis 59*(6), 2547–2581.

Barber, B. M. and T. Odean (2000a). Trading is hazardous to your wealth: The common stock investment performance of individual investors. *The journal of Finance 55*(2), 773–806.

Barber, B. M. and T. Odean (2000b). Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors. *The Journal of Finance 55*(2), 773–806.

Barber, B. M. and T. Odean (2002). Online investors: do the slow die first? *The Review of financial studies 15*(2), 455–488.

Barber, B. M. and T. Odean (2008). All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors. *Review of Financial Studies 21*(2), 785–818.

Barber, B. M., T. Odean, and N. Zhu (2008). Do retail trades move markets? *The Review of Financial Studies 22*(1), 151–186.

Barrot, J.-N., R. Kaniel, and D. Sraer (2016). Are retail traders compensated for providing liquidity? *Journal of Financial Economics 120*(1), 146–168.

Benartzi, S. and R. H. Thaler (2001). Naive diversification strategies in defined contribution saving plans. *American economic review 91*(1), 79–98.

Blume, M. E., D. B. Keim, et al. (2014). The changing nature of institutional stock investing. *Critical Finance Review 6*, 1–41.

Boehmer, E., C. M. Jones, X. Zhang, and X. Zhang (2021). Tracking retail investor activity. *The Journal of Finance 76*(5), 2249–2305.

Bretscher, L., L. Schmid, I. Sen, and V. Sharma (2025). Institutional Corporate Bond Pricing. *The Review of Financial Studies*, hhaf067.

Chang, Y.-C., H. Hong, and I. Liskovich (2015). Regression discontinuity and the price effects of stock market indexing. *The Review of Financial Studies 28*(1), 212–246.

Chaudhry, A. and J. Li (2025). Endogenous elasticities: Price multipliers are smaller for larger demand shocks.

Chen, J., H. Hong, and J. C. Stein (2001). Forecasting crashes: trading volume, past returns, and conditional skewness in stock prices. *Journal of Financial Economics 61*(3), 345–381.

Cremers, K. M. and A. Petajisto (2009). How active is your fund manager? A new measure that predicts performance. *The Review of Rinancial Studies 22*(9), 3329–3365.

Davis, C., M. Kargar, and J. Li (2023). Why is asset demand inelastic? *Why is Asset Demand Inelastic?: Davis, Carter/ uKargar, Mahyar/ uLi, Jiacui*.

Dhar, R. and N. Zhu (2006). Up close and personal: Investor sophistication and the disposition effect. *Management science 52*(5), 726–740.

Eaton, G. W., T. C. Green, B. S. Roseman, and Y. Wu (2022). Retail trader sophistication and stock market quality: Evidence from brokerage outages. *Journal of Financial Economics 146*(2), 502–528.

Fang, X., B. Hardy, and K. K. Lewis (2025). Who Holds Sovereign Debt and Why It Matters. *The Review of Financial Studies 38*(8), 2326–2361.

Foucault, T., D. Sraer, and D. J. Thesmar (2011). Individual investors and volatility. *The Journal of Finance 66*(4), 1369–1406.

Frazzini, A. and L. H. Pedersen (2014). Betting against beta. *Journal of Financial Economics 111*(1), 1–25.

Gabaix, X. and R. S. Koijen (2021). In search of the origins of financial fluctuations: The inelastic markets hypothesis. Technical report, National Bureau of Economic Research.

Gabaix, X., R. S. Koijen, F. Mainardi, S. S. Oh, and M. Yogo (2025). Limited risk transfer between investors: A new benchmark for macro-finance models. Technical report, National Bureau of Economic Research.

Glossner, S., P. Matos, S. Ramelli, and A. F. Wagner (2025). Do Institutional Investors Stabilize Equity Markets in Crisis Periods? Evidence from COVID-19. *Management Science*, mnsc.2022.03411.

Graves, D. (2025). What lies beneath zero: Censoring, demand estimation, and hidden beliefs. Working paper.

Greenwood, R., T. Laarits, and J. Wurgler (2023). Stock market stimulus. *The Review of Financial Studies 36*(10), 4082–4112.

Greenwood, R. and S. Nagel (2009). Inexperienced investors and bubbles. *Journal of Financial Economics 93*(2), 239–258.

Grossman, S. J. and M. H. Miller (1988). Liquidity and market structure. *the Journal of Finance 43*(3), 617–633.

Haddad, V., P. Huebner, and E. Loualiche (2025). How competitive is the stock market? theory, evidence from portfolios, and implications for the rise of passive investing. *American Economic Review 115*(3), 975–1018.

Haghighi, A., R. Faff, and B. Oliver (2024). Retail traders and co-movement: Evidence from Robinhood trading activity. *International Review of Financial Analysis 95*, 103431.

He, Z., B. Kelly, and A. Manela (2017). Intermediary asset pricing: New evidence from many asset classes. *Journal of Financial Economics 126*(1), 1–35.

Hendershott, T. and A. J. Menkveld (2014). Price pressures. *Journal of Financial Economics 114*(3), 405–423.

Ho, T. and H. R. Stoll (1981). Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics 9*(1), 47–73.

Huberman, G. (2001). Familiarity breeds investment. *The Review of Financial Studies 14*(3), 659–680.

Huebner, P. (2023). The Making of Momentum: A Demand-System Perspective. *SSRN Electronic Journal*.

Jansen, K. A. E. (2025). Long-Term Investors, Demand Shifts, and Yields. *The Review of Financial Studies 38*(1), 114–157.

Jiang, Z., R. J. Richmond, and T. Zhang (2024). A Portfolio Approach to Global Imbalances. *The Journal of Finance 79*(3), 2025–2076.

Kaniel, R., S. Liu, G. Saar, and S. Titman (2012). Individual Investor Trading and Return Patterns around Earnings Announcements. *The Journal of Finance 67*(2), 639–680.

Kaniel, R., G. Saar, and S. Titman (2008). Individual investor trading and stock returns. *The Journal of finance 63*(1), 273–310.

Kelley, E. K. and P. C. Tetlock (2013). How Wise Are Crowds? Insights from Retail Orders and Stock Returns. *The Journal of Finance 68*(3), 1229–1265.

Koijen, R. S. and M. Yogo (2019). A demand system approach to asset pricing. *Journal of Political Economy 127*(4), 1475–1515.

Koijen, R. S. J., R. J. Richmond, and M. Yogo (2024). Which Investors Matter for Equity Valuations and Expected Returns? *Review of Economic Studies 91*(4), 2387–2424.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica: Journal of the Econometric Society*, 1315–1335.

Laarits, T. and M. Sammon (2024). The retail habitat. *Available at SSRN 4262861*.

Lewellen, J. (2011). Institutional investors and the limits of arbitrage. *Journal of Financial Economics 102*(1), 62–80.

Li, F. (2025). Retail trading and asset prices: The role of changing social dynamics. *Available at SSRN 4236966*.

Li, Y., S. Sokolinski, and A. Tamoni (2024). Which Investors Drive Anomaly Returns and How? *SSRN Electronic Journal*.

Liang, Q., M. Najand, D. Selover, and L. Sun (2024). Retail Trading Around Earnings Announcements: Evidence from Robinhood Traders. *Journal of Behavioral Finance*, 1–24.

Lochstoer, L. A. and P. C. Tetlock (2020). What Drives Anomaly Returns? *The Journal of Finance 75*(3), 1417–1455.

Lou, D. (2012). A flow-based explanation for return predictability. *The Review of Financial Studies 25*(12), 3457–3489.

Lucas, R. E. (1976). Econometric policy evaluation: A critique. In *Carnegie-Rochester conference series on public policy*, pp. 19–46. North-Holland.

Mantel, P. and M. Sa. Democratizing or Demoralizing: The Impact of Robinhood on Trading Costs and Volatility.

Moss, A., J. P. Naughton, and C. Wang (2024). The Irrelevance of Environmental, Social, and Governance Disclosure to Retail Investors. *Management Science 70*(4), 2626–2644.

Odean, T. (1998). Are investors reluctant to realize their losses? *The Journal of finance 53*(5), 1775–1798.

Ozik, G., R. Sadka, and S. Shen (2021). Flattening the illiquidity curve: Retail trading during the covid-19 lockdown. *Journal of Financial and Quantitative Analysis 56*(7), 2356–2388.

Pagano, M. S., J. Sedunov, and R. Velthuis (2021). How did retail investors respond to the covid-19 pandemic? the effect of robinhood brokerage customers on market quality. *Finance Research Letters 43*, 101946.

Pavlova, A. and T. Sikorskaya (2023). Benchmarking intensity. *The Review of Financial Studies 36*(3), 859–903.

Pedersen, L. H. (2022). Game on: Social networks and markets. *Journal of Financial Economics 146*(3), 1097–1119.

Peress, J. and D. Schmidt (2020). Glued to the tv: Distracted noise traders and stock market liquidity. *The Journal of Finance 75*(2), 1083–1133.

Seasholes, M. S. and N. Zhu (2010). Individual investors and local bias. *The Journal of Finance 65*(5), 1987–2010.

Shleifer, A. (1986). Do demand curves for stocks slope down? *The Journal of Finance 41*(3), 579–590.

Stock, J. H. and M. Yogo (2002). Testing for weak instruments in linear IV regression. Technical report, National Bureau of Economic Research.

Vaarmets, T., K. Liivamägi, and T. Talpsepp (2019). How does learning and education help to overcome the disposition effect? *Review of Finance 23*(4), 801–830.

Welch, I. (2022). The wisdom of the robinhood crowd. *The Journal of Finance 77*(3), 1489–1527.

Wurgler, J. and E. Zhuravskaya (2002). Does arbitrage flatten demand curves for stocks? *The Journal of Business 75*(4), 583–608.

Appendices

A The Robinhood Portfolio

A Summary Statistics

Robintrack reports $\text{Acc}_t(n)$, i.e. the number of accounts holding each stock traded on Robinhood. Table A.1 reports summary statistics on $\text{Acc}_t(n)$.

Table A.1: **Robinhood Accounts Holding Summary Statistics**
This table reports summary statistics on the number of Robinhood accounts holding a stock $\text{Acc}_t(n)$ for all quarters in our data. The summary statistics are cross-sectional measures computed across all stocks within our investment universe. For example, the mean corresponds to $\frac{1}{N}\sum_{n=1}^{N}\text{Acc}_t(n)$. The number of observations corresponds to the number of stocks traded on Robinhood. Changes in $\text{Acc}_t(n)$ are computed for each stock-quarter pair.

	Mean	Std	Min	P25	Median	P75	Max	N(Obs.)
Accounts Holding $\text{Acc}_t(n)$								
2018 Q2	1,604	8,971	0	53	180	608	150,718	1,989
2018 Q3	1,881	10,945	0	57	207	704	180,324	2,020
2018 Q4	2,127	12,518	0	63	214	749	216,843	1,990
2019 Q1	2,408	13,523	0	75	259	908	261,099	2,104
2019 Q2	2,691	14,860	0	89	292	1,019	268,460	2,133
2019 Q3	2,853	15,671	0	100	322	1,088	297,496	2,177
2019 Q4	3,093	16,891	0	110	360	1,206	321,191	2,145
2020 Q1	5,794	29,902	0	167	550	1,983	649,148	2,216
2020 Q2	10,040	49,348	0	272	930	3,283	925,539	2,252
Changes in Accounts Holding $\text{Acc}_t(n) - \text{Acc}_{t-1}(n)$								
2018 Q3	312	2,693	-3,132	-6	9	59	69,294	1,765
2018 Q4	184	2,125	-19,706	-14	2	37	53,913	1,781
2019 Q1	337	2,382	-15,968	1	21	100	48,830	1,863
2019 Q2	238	1,932	-12,703	-10	7	68	40,444	1,928
2019 Q3	201	1,820	-17,337	-10	6	69	36,999	2,005
2019 Q4	220	2,297	-33,360	-14	5	60	47,560	2,017
2020 Q1	2,850	16,572	-4,394	18	102	577	327,957	2,117
2020 Q2	4,343	23,154	-11,138	52	226	1,025	431,398	2,152

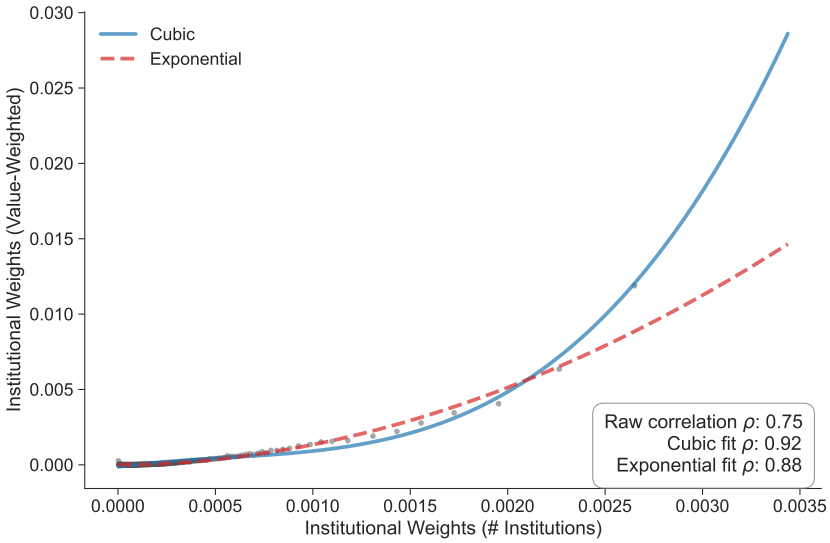
B Portfolio Weight Approximations

Our baseline approximation of the Robinhood portfolio, $w_t^{\text{RH}}(n) = \frac{\text{Acc}_t(n)}{\sum_{n=1}^{N} \text{Acc}_t(n)}$ follows Welch (2022), who finds that constructing this portfolio using data from Barber and Odean (2000b) is 97.1% correlated with the true equal-weighted household portfolio. This portfolio can be motivated as follows. Let $W_{a,t}$ denote the wealth of account a and $w_{a,t}(n)$ its portfolio weight in stock n. The aggregate Robinhood portfolio weight in stock n is

$$w_t^{\text{RH}}(n) = \frac{\sum_{a=1}^{\text{Acc}_t} W_{a,t} w_{a,t}(n)}{\sum_{a=1}^{\text{Acc}_t} W_{a,t}}. \tag{15}$$

If each account holding stock n represents the same dollar amount, then this trivially reduces to $w_t^{\text{RH}}(n) = \frac{\text{Acc}_t(n)}{\sum_{n=1}^{N} \text{Acc}_t(n)}$ as in (1). More simply, this approximation holds so long as the wealth distribution is uncorrelated with the users' investment universes, i.e., their sets of preferred stocks. Nevertheless, the precise mapping between account holders and portfolio holdings for Robinhood traders remains unknown. However, we can estimate the mapping from institutional investors, where both the number of holders $\text{Inst}_t(n)$ and the precise portfolio holdings are available. Figure A.1 illustrates the striking relationship between the aggregate institutional weights $w_t^{\text{Inst}}(n)$ and the number of holders $\text{Inst}_t(n)$.

Figure A.1: **Institutional Holders and Aggregate Portfolio Weights**
We plot the relationship between the number of institutional owners $\text{Inst}_t(n)$ and aggregate institutional portfolio weights. The blue line plots a cubic fit $f_{\text{poly}}(\cdot)$. The red dashed line plots an exponential fit $f_{\text{exp}}(\cdot)$.



We fit a third-order polynomial (blue line) and exponential function (red dashed line) to the relationship between institutional owners and aggregate institutional portfolio weights and denote the resulting mapping functions as $f_{\text{poly}}(\cdot)$ and $f_{\text{exp}}(\cdot)$. Then we apply the mapping functions to the number of Robinhood holders $\text{Acc}_t(n)$ to back out the implied Robinhood weight. Table A.2 plots the correlation of our baseline Robinhood portfolio $w_t^{\text{RH}}(n)$, with the alternative portfolios $w_t^{\text{RH,prc}}(n)$,$w_t^{\text{RH,poly}}(n)$, and $w_t^{\text{RH,exp}}(n)$.

Table A.2: **Correlation of Robinhood Portfolio Proxies**
The table plots the correlation between the baseline Robinhood portfolio $w_t^{\text{RH}}(n)$, the price-based Robinhood portfolio $w_t^{\text{RH,prc}}(n)$, the fitted Robinhood portfolio based on a cubic fit $w_t^{\text{RH,poly}}(n)$, and the fitted Robinhood portfolio based on an exponential fit $w_t^{\text{RH,exp}}(n)$.

Table A.3: Correlation Matrix of Robinhood Portfolio Weight Methods

	Equal	Price	Polynomial	Exponential
Equal	1.00	0.32	0.76	0.75
Price	0.32	1.00	0.30	0.30
Polynomial	0.76	0.30	1.00	0.99
Exponential	0.75	0.30	0.99	1.00

C Scaling the Implied Holdings

While the actual amount of shares held by Robinhood traders $Q_t^{\text{RH}}(n)$ is unobserved, we reverse-engineer an approximation using (1) and Robinhood's reported equities assets under management from their S-1 filing from July 1st, 2021 and their August 18th, 2021 10-Qs,

$$Q_t^{\text{RH}}(n) = \frac{w_t^{\text{RH}}(n) A_t^{\text{RH}}}{P_t(n)}. \tag{16}$$

The S-1 filing reports equity assets under management for December 31st, 2019, as well as March 31st, 2020. The August 18th 10-Q reports assets under custody for June 30th, 2020. In order to infer the shares held in Q2, we need to approximate Robinhood's AUM given the reported figures. In order to get AUM approximations for all quarters before December 2019 that are consistent the evolution of open positions, we fit a polynomial to $\frac{1}{N}\sum_{n=1}^{N} \text{Acc}_t(n)$ and evaluate the function at different dates, given that in December 31st it took $11.7 billion. The

residual shares not held by Robinhood traders or institutions make up the household sector. In the rare cases, where institutional and implied Robinhood holdings exceed the shares outstanding, we rescale Robinhood traders' shares held to be $\tilde{Q}_t^{\mathrm{RH}}(n) = S_t - \sum_{i=1}^{I} Q_t^i(n)$. Robinhood portfolio weights are then recomputed as $w_t^{\mathrm{RH}}(n) = \frac{\tilde{Q}_t^{\mathrm{RH}}(n) P_t(n)}{\sum_{n=1}^{N} \tilde{Q}_t^{\mathrm{RH}}(n) P_t(n)}$.

B Institutional Holdings

A Institutional Investors

We obtain quarterly institutional holdings data on stocks traded on the New York Stock Exchange (NYSE), the NYSE American, and the NASDAQ from Refinitiv. We restrict our sample to holdings disclosed in 13F filings. Note that Refinitiv Ownership reports holdings by investment managers. An investment manager does not always correspond to an entire institution, but can be a subsidiary.[20] Using Refinitiv, we identify the parent institution of each investment manager and aggregate holdings at the parent institution level.

Table B.1: **Institutional Ownership Data Summary**
This table reports summary statistics on institutional holdings. The statistics displayed are the time series average of the June 2018 to June 2020 quarterly values.

	Number of FI	Percentage of Market Held	Assets under Management		Number of Stocks Held		Number of Stocks in Universe	
			Median	90th Pct	Median	90th Pct	Median	90th Pct
Banks	142	6	454	29,369	157	1,087	303	1,949
Insurance Companies	22	1	1,115	44,865	112	580	196	923
Pension Funds	42	4	9,937	50,643	738	1,917	1,017	2,442
Investment Advisors	1,543	40	398	6,677	85	441	151	865
Hedge Funds	325	2	382	3,119	33	173	82	569
Other	30	1	686	10,599	72	1,102	110	2,579

[20] For example, BlackRock is split between 16 investment managers.

We compute the percentage of ownership as the number of shares held divided by shares outstanding. We winsorize ownership shares above 50% and proportionally scale down holdings so that total institutional ownership does not exceed 100%.[21] We construct each institution's equity portfolio weights as the dollar holdings in each stock (price times shares held) divided by their assets under management. We compute assets under management as the sum of an institution's dollar holdings. Following KY (2019), we attribute the holdings of institutions with less than $10 million under management to the household sector. Similarly, all institutions that do not have any holdings in any inside assets or the outside asset are merged with households.

We regroup Refinitiv's 14 investor categories into six investor types. Insurance companies and hedge funds remain as they are. Banks and trusts are grouped with holding companies, investment advisors with investment advisors/hedge funds, and pension funds with sovereign wealth funds. Finally, endowment funds, venture capital firms, research firms, foundations, private equity firms, and corporations are grouped together (labelled as *Other*). Table B.1 reports summary statistics on institutional holdings by investor type from June 2018 to June 2020.

B Sparse Portfolios

Estimating the demand system requires estimating demand coefficients for each individual investor. However, most of the institutions in our sample hold very concentrated portfolios or have filed few 13F filings. Indeed, the majority of institution-quarter pairs report less than 100 holdings. In order to obtain sufficiently many observations for the demand estimation, we pool institutions with insufficient holdings together and estimate demand coefficients for the

[21] Lewellen (2011) finds that institutional holdings exceeding 100% of shares outstanding are not data errors but the result of double-counting shares that were short-sold.

pooled entities. We establish two cutoffs; i) a cross-sectional cutoff of 100 positive holdings in our investment universe (excluding the outside asset) that must be satisfied in every quarter, and ii) a time series cutoff of 12 quarters of available 13F filings. Institutions that do not meet the required cutoffs are only pooled together with institutions of the same investor type. Additionally, investment advisors are pooled with investment advisors that share the same investment style. We obtain advisors' investment style from Refinitiv. Within each type (banks, hedge funds, insurance companies, pension funds, and other) and investment style (core value, core growth, growth at a reasonable price (GARP), growth, hedge fund, deep value, income value, yield, and other), we sort institutions by quantiles of average AUM. The holdings of the institutions belonging to the same type/style and quantile are summed together and form the aggregated holdings of our pooled entities. The number of quantiles is set such that the cross-sectional and the time series cutoffs are satisfied for every pool. Note that, to conform with our panel approach, pools are immutable. Thus, an institution is uniquely assigned to the same pool in every quarter.

C Proofs

C.1 Proof of Equation (14)

Recall the market clearing condition, which implies that $\mathbf{p}_t = \log\Big(\sum_j A_t^j \mathbf{w}_t^j(\mathbf{p}_t)\Big) - \mathbf{s}_t$. Differentiating both sides with respect to the log latent demand of Robinhood traders $\log(\boldsymbol{\epsilon}_t^{\text{RH}})$ yields

$$\frac{\partial \mathbf{p}_t}{\partial \log(\boldsymbol{\epsilon}_t^{\text{RH}})'} = \mathbf{H}_t^{-1}\left(\sum_j A_t^j \frac{\partial \mathbf{w}_t^j}{\partial \mathbf{p}_t'} \frac{\partial \mathbf{p}_t}{\partial \log(\boldsymbol{\epsilon}_t^{\text{RH}})'} + A_t^{\text{RH}} \frac{\partial \mathbf{w}_t^j}{\partial \log(\boldsymbol{\epsilon}_t^{\text{RH}})'}\right), \tag{17}$$

where $\mathbf{H}_t = \sum_j A_t^j \text{diag}(\mathbf{w}_t^j)$. Solving for $\frac{\partial \mathbf{p}_t}{\partial \log(\boldsymbol{\epsilon}_t^{\text{RH}})'}$ yields

$$\frac{\partial \mathbf{p}_t}{\partial \log(\boldsymbol{\epsilon}_t^{\text{RH}})'} = \left(\mathbf{I} - \sum_j A_t^j \mathbf{H}_t^{-1} \frac{\partial \mathbf{w}_t^j}{\partial \mathbf{p}_t'}\right) A_t^{\text{RH}} \mathbf{H}_t^{-1} \frac{\partial \mathbf{w}_t^{\text{RH}}}{\partial \boldsymbol{\epsilon}_t^{\text{RH}\prime}}, \tag{18}$$

where the last term is given by

$$\frac{\partial \mathbf{w}_t^{\text{RH}}}{\partial \log(\boldsymbol{\epsilon}_t^{\text{RH}})'} = \text{diag}(\mathbf{w}_t^{\text{RH}}) - \mathbf{w}_t^{\text{RH}} \mathbf{w}_t^{\text{RH}\prime}. \tag{19}$$

Substituting (**??**) and (19) in (18) yields

$$\frac{\partial \mathbf{p}_t}{\partial \log(\boldsymbol{\epsilon}_t^{\text{RH}})'} = \left(\mathbf{I} - \sum_j A_t^j \gamma^j \mathbf{H}_t^{-1} \mathbf{G}_t^j\right)^{-1} A_t^{\text{RH}} \mathbf{H}_t^{-1} \mathbf{G}_t^{\text{RH}}, \tag{20}$$

where $\mathbf{H}_t = \sum_j A_t^j \text{diag}(\mathbf{w}_t^j)$. The matrix inside the inverse captures the market's demand elasticity, which determines the amplification of demand shocks across the stock universe. If $\text{diag}(\mathbf{w}_j) - \mathbf{w}_j \mathbf{w}_j' \approx \text{diag}(\mathbf{w}_j)$, then $A_t^j \mathbf{H}_t^{-1} \mathbf{G}_t^j = \text{diag}(\mathbf{z}_t^j)$, where $z_t^j(n) = \frac{w_t^j(n) A_t^j}{\sum_k w_t^k(n) A_t^k}$ is the ownership share of j in stock n. Thus, we can write

$$\frac{\partial \mathbf{p}_t}{\partial \log(\boldsymbol{\epsilon}_t^{\text{RH}})'} = \left(\mathbf{I} - \sum_j \gamma^j \text{diag}(\mathbf{z}_t^j)\right)^{-1} \text{diag}(\mathbf{z}_t^{\text{RH}}). \tag{21}$$

D Supplementary Plots and Tables

Table D.1: **Estimated Demand Functions: Pre-2020**
The table reports the estimated demand coefficients for Robinhood traders, households, and 13F institutions. The sample period is 2018 to 2019. We group 13F institutions into 6 groups based on Refinitiv's type labels (hedge funds, banks, insurance companies, investment advisors, pension funds) and report AUM-weighted average for each group.

		Estimated Coefficients						
	Wealth Share	γ^j	β^j_{be}	$\beta^j_{\text{inv.}}$	β^j_{profit}	β^j_{beta}	$\beta^j_{\text{div/be}}$	Demand Elasticity
Institutional Investors								
Pension Fund	4%	0.98	16.97	-2.14	0.15	-2.78	0.28	**0.03**
Bank	6%	0.98	32.95	0.02	0.12	-5.08	0.20	**0.04**
Investment Advisor	48%	0.91	27.86	3.29	0.27	0.05	-0.01	**0.10**
Other	1.0%	0.86	50.45	3.12	0.44	-0.27	0.03	**0.16**
Insurance Company	2%	0.83	-0.17	4.03	-0.54	-4.62	0.11	**0.20**
Hedge Funds	2%	0.67	0.57	7.78	0.67	6.44	-0.07	**0.35**
Households	37%	0.81	3.45	0.33	0.16	-5.80	0.02	**0.20**
Robinhood Traders	0.03%	0.22	-18.77	8.77	-1.23	20.97	4.15	**0.78**

Table D.2: **Attention-Induced Latent Demand: Robustness**
The table reports simple panel regressions of Robinhood traders' latent demand $\varepsilon^i_t(n)$ onto quarterly lagged absolute returns $|r_{t-1}|$ and quarterly lagged trading volume. Each column corresponds to a different Robinhood portfolio: The baseline Robinhood portfolio $w^{\text{RH}}_t(n)$, the price-weighted Robinhood portfolio $w^{\text{RH,prc}}_t(n)$, the fitted Robinhood portfolio based on a cubic fit $w^{\text{RH,poly}}_t(n)$, and the fitted Robinhood portfolio based on an exponential fit $w^{\text{RH,exp}}_t(n)$.

	Baseline w^{RH}_t	Price-weighted $w^{\text{RH,prc}}_t$	Polynomial $w^{\text{RH,poly}}_t$	Exponential $w^{\text{RH,exp}}_t$
	(1)	(2)	(3)	(4)
$\lvert r_{t-1}\rvert$	1.270**	0.812**	0.576***	2.311**
	(3.54)	(3.46)	(5.70)	(3.48)
Volume_{t-1} (millions)	0.194***	0.107***	0.124***	0.330***
	(19.21)	(15.36)	(20.93)	(16.53)
Time Fixed Effects	Y	Y	Y	Y
R^2 Within	0.251	0.122	0.192	0.212
Observations	9,810	9,810	2,967	9,832